

Moments of random multiplicative functions with polynomial coefficients

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Abstract

Let f be a Steinhaus random multiplicative function and let g be a polynomial of degree d . Write

$$S_N = \frac{1}{\sqrt{N}} \sum_{n \leq N} f(n) e(g(n)).$$

We prove a quantitative dichotomy for the moments of S_N : for each integer $s \geq 2$, either $\mathbb{E}|S_N|^{2s}$ is close to the Gaussian moment $s!$, or the coefficients of g can be approximated by rationals with a small denominator. In particular, if the coefficients of g satisfy a Diophantine condition, then S_N converges in law to a complex normal distribution with mean 0 and variance 1. Lean 4 code for the proofs is provided, for convenience of verification.

1 Introduction

A Steinhaus random multiplicative function is a probabilistic model for the Möbius function. The definition is as follows. Let $\{f(p)\}_{p \text{ prime}}$ be a sequence of independent random variables, identically uniformly distributed on the complex unit circle. We then extend the definition of f to all natural numbers by complete multiplicativity. That is, writing $p^k \parallel n$ to mean that p^k is the highest power of p that divides n , we have

$$f(n) := \prod_{p^k \parallel n} f(p)^k.$$

As in the definition, the sequence $f(n)$ exhibits both randomness and multiplicativity. Hence one may ask how the random multiplicative structure of $f(n)$ interacts with the structure of a deterministic sequence? The sums of interest are $\sum_{n \leq N} f(n)w(n)$, for a deterministic sequence w . In this paper, we take $w(n) = e(g(n))$, where $e(x) = \exp(2\pi i x)$ and g is a real polynomial of degree d ,

$$g(n) = \sum_{j=0}^d \beta_j n^j,$$

where $\beta_j \in \mathbb{R}$, $j = 0, 1, \dots, d$ are the coefficients of the polynomial. Write

$$S_N = \frac{1}{\sqrt{N}} \sum_{n \leq N} f(n) e(g(n)).$$

Here $\sqrt{N}$ is the size of the standard deviation of $\sum_{n \leq N} f(n) e(g(n))$.

We are interested in the moments of the random variables S_N . We give a quantitative dichotomy for the moments of S_N — the diophantine property of the coefficients β_i determines the moment behaviour.

We adapt the definition of the smoothness norm as in [3] to describe the Diophantine property. Throughout, $\|\cdot\|_{\mathbb{R}/\mathbb{Z}}$ denotes distance to the nearest integer.

Definition 1 ($C^\infty[N]$ norm). For $g(n) = \sum_{j=0}^d \beta_j n^j$ as above and $N \geq 1$, we use the monomial-coefficient smoothness norm

$$\|g\|_{C^\infty[N]} := \max_{1 \leq j \leq d} N^j \|\beta_j\|_{\mathbb{R}/\mathbb{Z}}.$$

As in Green–Tao [3, Definition 2.7], this smoothness norm is designed to capture the notion of a polynomial sequence which is slowly-varying. Note that an upper bound on $\|qg\|_{C^\infty[N]}$ for some small nonzero integer q says that the coefficients of g can be approximated by rationals with a small denominator.

The following theorem is our main result.

Theorem 2 (Moment dichotomy). *Given integers $d \geq 1$ and $s \geq 2$, there exists a constant $c_0 = c_0(d, s) > 0$ such that the following holds. For all integers N large enough depending only on d and s , let S_N be defined as above for any polynomial g of degree d . For every $N^{-c_0} < \delta < 1/8$, at least one of the following holds:*

1.

$$\left| \mathbb{E} |S_N|^{2s} - s! \right| \ll_{d,s} \delta (\log N)^{O(s^2)}.$$

2. *There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that*

$$\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}.$$

Note that $s!$ is exactly the $2s$ -th Gaussian moment of a complex normal distribution with mean 0 and variance 1. In other words, either the $2s$ -th moment is close to $s!$, or qg is slowly varying for some small nonzero integer q .

Let us note that allowing the second alternative is necessary. If we take g to be a constant, then $|S_N| = |N^{-1/2} \sum_{n \leq N} f(n)|$, and the $2s$ -th moment of this sum is of size $(\log N)^{\Theta(s^2)}$ rather than $s!$, by Harper, Nikeghbali and Radziwiłł [6] and Heap and Lindqvist [7].

When $s = 1$ one has $\mathbb{E} |S_N|^2 = 1$ immediately, by Steinhaus orthogonality, that is, since $\mathbb{E} f(p) = 0$ and natural numbers have unique prime factorisations, it is immediate that for all natural numbers n and m ,

$$\mathbb{E} [f(n) \overline{f(m)}] = \mathbf{1}_{n=m}.$$

Let us also note that the odd moments of S_N tend to zero as $N \rightarrow \infty$, which can be shown by easy arguments. See the arguments in the proof of Corollary 3.

For related results regarding the moments of partial sums of random multiplicative functions, Pandey, Wang and Xu [8] showed that the $2s$ -th moments of $H^{-1/2} \sum_{x < n \leq x+H} f(n)$ match the Gaussian values $s!$ when H is a sufficiently short interval. Wang and Xu [12] showed that the $2s$ -th moments of $N^{-1/2} \sum_{n \leq N} f(P(n))$ match the Gaussian values $s!$, for a polynomial $P \in \mathbb{Z}[x]$ with at least two distinct complex roots. Meanwhile, Benatar, Nishry and Rodgers [1] showed that

$$\mathbb{E} \left| \int_0^1 \left| N^{-1/2} \sum_{n \leq N} f(n) e(n\theta) \right|^{2s} d\theta - s! \right|^2 \rightarrow 0,$$

where the expectation is over f . This holds for $s \ll (\log N / \log \log N)^{1/3}$. Each of the results above yields a central limit theorem.

If the coefficients of g satisfy a Diophantine condition, then the following central limit theorem holds.

Corollary 3 (Central limit theorem). *Write $g(n) = \beta_d n^d + \cdots + \beta_1 n + \beta_0$. Assume that for every $\varepsilon > 0$ there exists $C = C(g, \varepsilon) > 0$ such that for every nonzero integer q ,*

$$\max_{1 \leq i \leq d} \|q\beta_i\|_{\mathbb{R}/\mathbb{Z}} \geq C \exp(-|q|^\varepsilon).$$

Then S_N converges in law to a complex normal distribution with mean 0 and variance 1.

For example, the hypothesis holds if some coefficient β_i is an algebraic irrational. In fact, it can be shown that the exceptional set has Hausdorff dimension zero.

From Harper [5] it follows that Corollary 3 fails for rational phases, so some Diophantine condition is necessary. Previously, Soundararajan and Xu [10] gave a general criterion on a deterministic weight w under which $N^{-1/2} \sum_{n \leq N} f(n)w(n)$ converges in law to a centred complex normal of variance 1. Their proof uses the martingale central limit theorem and a fourth-moment computation. For the linear phase $w(n) = e(n\theta)$, [10, Theorem 1.6] imposes a condition of the same shape with a single value $\varepsilon = 1/50$. Here, however, the Diophantine condition is assumed for every $\varepsilon > 0$. This is because we use the moment method to prove the central limit theorem, and ε must be taken smaller as the moment order increases in order to make sure the higher moments of S_N tend to Gaussian moments. But in fact, for the fourth-moment, we only need to assume the Diophantine condition for a fixed ε depending only on the degree of the polynomial.

The implied constants in Theorem 2 depend on s , and we state the result for each fixed s . The same argument should allow s to grow slowly with N , once its dependence is tracked uniformly. For instance, one expects this for $s = o((\log N / \log \log N)^{1/3})$.

We believe that Theorem 2 can be extended to other sequences, for example nilsequences such as $w(n) = e(\{n\alpha\}n\beta)$ where $\alpha, \beta \in \mathbb{R}/\mathbb{Z}$, and $\{n\alpha\}$ is the fractional part of $n\alpha$. But the central limit theorem need not hold for an arbitrary Lipschitz function F in place of $e(\cdot)$. See Schlitt [9] for a necessary and sufficient condition on F under which $\sum_{n \leq N} f(n)F(n\alpha)$ obeys a central limit theorem, where F is 1-periodic of bounded variation and α is irrational.

The main input of Theorem 2 is the quantitative Weyl theorem of Green–Tao [3] for polynomial orbits on the circle. With this main input, the other ingredients include the inclusion-exclusion decomposition, some parameterisation lemmas, divisor sum estimates and effective recurrence arguments that appeared in [3] and [2, §3]. The proof strategy is further explained in Section 3.1.

1.1 Formalisation

For convenience of verification, we provide Lean 4 code for the proofs of the results at

https://github.com/cutedonkeyhhh/RMF_repo.

See this repository for the details.

1.2 Notation

We write $f \ll g$, or equivalently $f = O(g)$, if $|f| \leq C|g|$ for an absolute constant C . Subscripts indicate the dependence of implied constants on parameters.

The k -fold divisor function is denoted τ_k . If $\mathcal{B}$ is a set and $\mathcal{P}$ is a property, we write $\mathcal{B} \cap \{\mathcal{P}\}$ for the subset of $\mathcal{B}$ consisting of those elements that satisfy $\mathcal{P}$.

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2 Setup and inductive statement

The proof of Theorem 2 is by induction on s . We begin with the divisor bounds, which will be used throughout the paper, all of which are standard.

Lemma 4 (Divisor sums). *For integers $N \geq 3, l \geq 1$, and the l -fold divisor function τ_l , we have*

$$\sum_{n \leq N} \tau_l(n) \leq N(1 + \log N)^{l-1} \leq N(2 \log N)^{l-1},$$

$$\sum_{n \leq N} \tau_l(n)^2 \leq N(2 \log N)^{l^2-1},$$

and for any positive integer $A < N$,

$$\sum_{A \leq n \leq N} \frac{\tau_l(n)^2}{n^2} \leq \frac{5(2 \log N)^{l^2-1}}{A}.$$

Moreover,

$$\sum_{n \leq N} \frac{\tau_l(n)}{n} \leq (2 \log N)^l.$$

Proof. The first inequality is [1, Lemma 3.1], and the second is [1, (3.4)]. The third follows from the second by summation by parts. The last is $\sum_{n \leq N} \tau_l(n)/n \leq (\sum_{n \leq N} n^{-1})^l \leq (1 + \log N)^l \leq (2 \log N)^l$. $\square$

2.1 Normalised and unnormalised moments

We now consider how to compute the $2s$ -th moment of the partial sum S_N . Set

$$U_s(N) = \mathbb{E} \left| \sum_{n \leq N} f(n) e(g(n)) \right|^{2s}, \quad \mathcal{M}_s(N) = N^{-s} U_s(N) = \mathbb{E} |S_N|^{2s},$$

where $\mathcal{M}_s(N)$ is the normalised moment. It is a natural idea to transform the problem of computing the $2s$ -th moment of S_N to the computation of the exponential sum over some structured set, which is a purely arithmetic problem. For $N \geq 1$, set

$$V = \{(n_1, \dots, n_s, m_1, \dots, m_s) \in [1, N]^{2s} \cap \mathbb{Z}^{2s} : n_1 \cdots n_s = m_1 \cdots m_s\}.$$

This set V is the playground. Moreover, we need the following notation. For any subset $\mathcal{F} \subseteq V$, denote by $S_g(\mathcal{F})$ the sum of the exponential function over $\mathcal{F}$, that is,

$$S_g(\mathcal{F}) = \sum_{(\vec{n}_s, \vec{m}_s) \in \mathcal{F}} \prod_{i=1}^s e(g(n_i)) \overline{e(g(m_i))}.$$

With the above notation, we have the following moment formula.

Lemma 5 (Moment formula). $U_s(N) = S_g(V)$.

Proof. This identity is a direct consequence of the orthogonality of f , that is, for all natural numbers n and m ,

$$\mathbb{E}[f(n)\overline{f(m)}] = \mathbf{1}_{n=m}.$$

□

This lemma says that the computation of the 2st moment of the partial sum can be reduced to the computation of the exponential sum over the set V . Hence the structure of the set V is crucial. We will study the structure of V in the next section.

We prove Theorem 2 by induction on s . When $s = 1$, the condition defining V is simply $n_1 = m_1$. Lemma 5 therefore gives $U_1(N) = N$ and $\mathcal{M}_1(N) = 1$. Write $\mathcal{H}(1)$ for this identity. This is the base of the induction. Next we give the inductive statement.

2.2 The inductive statement

For an integer $s \geq 2$, write $\mathcal{H}(s)$ for the following proposition. For every integer $d \geq 1$, there exists a constant $c_0 = c_0(d, s) > 0$ such that the following holds. For every polynomial g of degree d , for all integers N large enough depending only on d and s , and for every $N^{-c_0} < \delta < 1/8$, at least one of the following holds:

1.

$$|\mathcal{M}_s(N) - s!| \ll_{d,s} \delta (\log N)^{O(s^2)}.$$

2. There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that

$$\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}.$$

Thus $\mathcal{H}(s)$ is exactly the conclusion of Theorem 2 at moment order $2s$.

3 The inclusion–exclusion decomposition

From the previous section, we have reduced the problem of computing the 2st moment of the partial sum S_N to the computation of the exponential sum over the set V . We now describe the set V and the inclusion–exclusion decomposition. Firstly, it is easy to see that the solution set V of the diophantine equation

$$n_1 \cdots n_s = m_1 \cdots m_s$$

contains the trivial solutions, in which $\vec{m}_s$ is a rearrangement of $\vec{n}_s$: that is, $n_i = m_{\sigma(i)}$ for $i = 1, \dots, s$, for some $\sigma \in S_s$, which is the symmetric group on s letters. On each such point, reindexing the product over m_i gives

$$\prod_{i=1}^s e(g(n_i)) \overline{e(g(m_i))} = \prod_{i=1}^s e(g(n_i)) \overline{e(g(m_{\sigma(i)}))} = \prod_{i=1}^s e(g(n_i)) \overline{e(g(n_i))} = 1,$$

so each trivial solution contributes 1 to $U_s(N)$. It can be shown that the number of distinct trivial solutions is $s!N^s + O_s(N^{s-1})$. Since each contributes 1, this is the main term of $U_s(N)$.

It is hard to directly compute the exponential sum over the non-trivial solutions in V . This forces us to study the structure of V .

The set V exhibits a lot of symmetry, since it is invariant under the action of the symmetric group S_s on the indices $1, \dots, s$. The first step of this work is to decompose the set V into structured sets by making use of this symmetry. We now describe this decomposition.

Let $A > 0$ (to be chosen later). For $i = 1, \dots, s$, define

$$\mathcal{F}_i = \{(\vec{n}_s, \vec{m}_s) \in V : \gcd(n_1, m_i) \geq A\}.$$

We let $\mathcal{F}_i$ implicitly depend on the parameter A . The above definition loosely captures the information of the greatest common divisors of n_1 and m_i as i varies. The reason we require the greatest common divisor to be larger than A is to guarantee that the effective recurrence argument of [2, §3] can be applied. Each $\mathcal{F}_i$ contains the solutions where $n_1 = m_i$, which will contribute to the main term. The structure is easier when we consider the exponential sum over each $\mathcal{F}_i$ separately. We are able to approximate the whole solution set V by the union of the $\mathcal{F}_i$ up to a negligible sparse complement, which is the content of the following lemma.

Lemma 6 (Sparse complement). *Let $s \geq 2$ and $N \geq 3$ be integers, and let A be a positive integer. With V and $\mathcal{F}_i$ as above,*

$$\#(V \setminus \cup_{i=1}^s \mathcal{F}_i) \leq N^{s-1} A^{2s} (2s \log N)^{(s+1)^2}.$$

Proof. If $(n_1, \dots, n_s, m_1, \dots, m_s) \in V \setminus \cup_i \mathcal{F}_i$, then $\gcd(n_1, m_j) < A$ for all $j = 1, \dots, s$. Since $n_1 \mid m_1 \cdots m_s$, one has $n_1 \mid \prod_{j=1}^s \gcd(n_1, m_j)$, hence $n_1 \leq A^s$.

Fix $n_1 = g$. The equation $n_1 \cdots n_s = m_1 \cdots m_s$ is equivalent to $m_1 \cdots m_s = gn$ for some $1 \leq n \leq N^{s-1}$ (writing $n = n_2 \cdots n_s$). For each such n , the number of $(n_2, \dots, n_s, m_1, \dots, m_s)$ with $n_2 \cdots n_s = n$ and $m_1 \cdots m_s = gn$ is at most $\tau_s(gn) \tau_{s-1}(n)$. Thus the fibre over $n_1 = g$ has size at most

$$\sum_{n \leq N^{s-1}} \tau_s(gn) \tau_{s-1}(n).$$

By Cauchy–Schwarz and the square-sum bound in Lemma 4, together with $\log(gN^{s-1}) \leq s \log N$ for $g \leq N$,

$$\sum_{n \leq N^{s-1}} \tau_s(gn) \tau_{s-1}(n) \leq gN^{s-1} (2s \log N)^{s^2}.$$

Summing over $g \leq A^s$ and using $s^2 \leq (s+1)^2$ gives

$$\#(V \setminus \cup_i \mathcal{F}_i) \leq \sum_{g \leq A^s} gN^{s-1} (2s \log N)^{s^2} \leq N^{s-1} A^{2s} (2s \log N)^{(s+1)^2}.$$

□

By inclusion–exclusion,

$$\sum_{\cup_i \mathcal{F}_i} = \sum_i \sum_{\mathcal{F}_i} - \sum_{i < j} \sum_{\mathcal{F}_i \cap \mathcal{F}_j} + \cdots + (-1)^{s-1} \sum_{\mathcal{F}_1 \cap \cdots \cap \mathcal{F}_s}. \quad (1)$$

Hence summation over the union $\cup_i \mathcal{F}_i$ can be decomposed into the sum over the single sets $\mathcal{F}_i$ and the sum over the intersections $\mathcal{F}_{i_1} \cap \cdots \cap \mathcal{F}_{i_k}$. Next we explain the strategy to prove Theorem 2.

3.1 Proof strategy

We prove Theorem 2 by induction on $s \geq 2$. Lemma 6 shows that the complement of $\bigcup_i \mathcal{F}_i$ in V is negligible, so the sum over V reduces to the sum over the union. By the inclusion–exclusion above, this sum splits into the single sets $\mathcal{F}_i$ and the intersections $\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_k}$. In Section 5, we estimate the single sets, and in Section 6, we estimate the intersections.

For both estimates, it is convenient to separate the diagonal and off-diagonal parts. For a single set $\mathcal{F}_i$, the diagonal part is the subset on which $n_1 = m_i$, and the off-diagonal part is

$$\mathcal{F}_i^\circ := \mathcal{F}_i \setminus \{n_1 = m_i\}.$$

For an intersection $\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_k}$, the diagonal part is the subset on which $n_1 = m_{i_1}$, and the off-diagonal part is

$$(\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_k})^\circ := (\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_k}) \setminus \{n_1 = m_{i_1}\}.$$

Figure 1 records these two off-diagonal sets.

The main term comes from the diagonal part of each $\mathcal{F}_i$, by the inductive hypothesis $\mathcal{H}(s-1)$, and the rest contributes no main term. The diagonal part of each intersection is controlled by the divisor bound. The off-diagonal parts of both $\mathcal{F}_i$ and the intersections are explained below, and this is the most technical part of the paper.

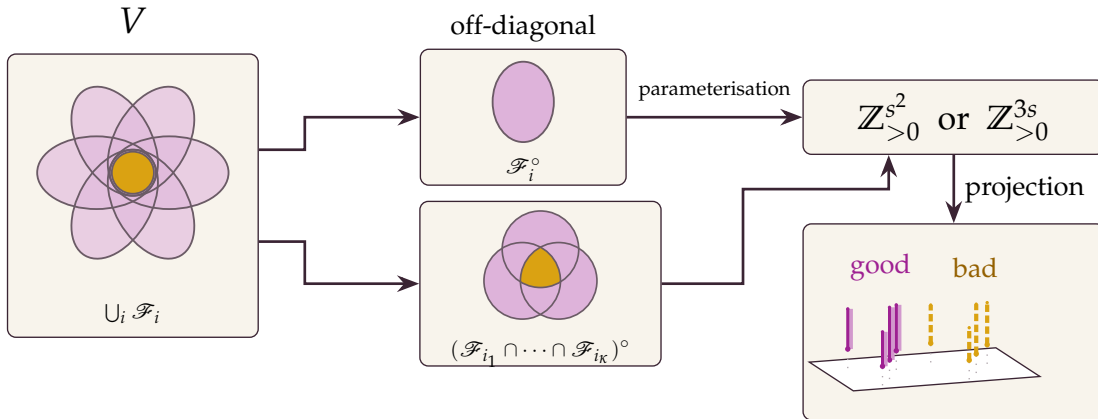


Figure 1: The proof strategy

The two off-diagonal sets in Figure 1 are treated by the same sequence of steps. Parameterisation lemmas place the solutions in $\mathbb{Z}_{>0}^{s^2}$ or in $\mathbb{Z}_{>0}^{3s}$. A projection then forgets some of these coordinates, leaving a base together with a fibre. The base points are split into good and bad according to the size of the exponential sum over the fibre. The number of bad base points is controlled by the Diophantine dichotomy of Green–Tao, explained in the next section.

4 Diophantine dichotomy

This section explains the diophantine dichotomy result needed for the proofs. We prove Lemma 10 following the argument of [2, §3]. The main input is the quantitative Leibman theorem of Green–Tao [3, Theorem 2.9, Proposition 4.3], specialised to polynomials on the circle, which is a quantitative Weyl theorem.

Our notion of smoothness norm is slightly different from the one used in Green–Tao. They [3, Definition 2.7] use the binomial basis and the norm

$$g(n) = \sum_{j=0}^d \alpha_j \binom{n}{j}, \quad \|g\|_{C_{\text{GT}}^\infty[N]} := \sup_{1 \leq j \leq d} N^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}.$$

By a change of basis, there is an integer q_d depending only on d such that

$$\|q_d g\|_{C^\infty[N]} \ll_d \|g\|_{C_{\text{GT}}^\infty[N]}.$$

So the following statement does not change if we use the monomial basis instead.

Theorem 7 (Green–Tao [3, Theorem 2.9, Proposition 4.3], for polynomials on the circle). *Let $p: \mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be a polynomial of degree at most d , let $M \geq 1$, and let $0 < \delta < 1/2$. Then at least one of the following holds:*

1. *The sequence $(p(n))_{n=1}^M$ is δ -equidistributed in $\mathbb{R}/\mathbb{Z}$, in the sense that*

$$\left| \frac{1}{M} \sum_{n=1}^M F(p(n)) - \int_{\mathbb{R}/\mathbb{Z}} F \right| \leq \delta \|F\|_{\text{Lip}}$$

for every Lipschitz function $F: \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{C}$.

2. *There exists a nonzero integer k such that*

$$|k| \ll_d \delta^{-O_d(1)}, \quad \|kp\|_{C^\infty[M]} \ll_d \delta^{-O_d(1)}.$$

This dichotomy says that, either the polynomial sequence is equidistributed or the polynomial is non-diophantine. Moreover, this result is quantitative, making it convenient for our application.

Several more ingredients from their work are used. We will also use the following effective recurrence lemma.

Lemma 8 (Effective recurrence, Green–Tao [3, Lemma 3.2]). *Let $\alpha \in \mathbb{R}$, let $0 < \sigma < 1/2$, and let $0 < \varepsilon \leq \sigma/2$. Let $I \subseteq \mathbb{R}/\mathbb{Z}$ be an interval of length ε , and let $N \geq 1$. Suppose that $\alpha n \in I$ for at least σN integers n with $1 \leq n \leq N$. Then there exists $k \in \mathbb{Z}$ with $0 < |k| \ll \sigma^{-O(1)}$ such that*

$$\|k\alpha\|_{\mathbb{R}/\mathbb{Z}} \ll \varepsilon \sigma^{-O(1)} / N.$$

We will also use the following Waring-type count.

Lemma 9 (Green–Tao [2, Lemma 3.3]). *Let $j \geq 1$ and $K \geq 1$ be integers, let $0 < \eta \leq 1$, and let $S \subseteq [K]$ with $|S| = \eta K$. Let $t \geq 2^j + 1$. Then $\gg_{j,t} \eta^{2t} K^j$ integers in the interval $[tK^j]$ can be written in the form $k_1^j + \dots + k_t^j$ with $k_1, \dots, k_t \in S$.*

With these inputs, we prove the next lemma formulated in this way for various adaptations.

Lemma 10 (Exponential dichotomy). *There exists a constant $C_d > 5$, depending only on d , with the following property. Let g be a real polynomial of degree d . Let $0 < \delta < 1/8$, let $0 < D < N$, and put $N_D = \lfloor N/D \rfloor$. Suppose that*

$$\frac{N}{D} > \delta^{-C_d},$$

and let $L = [a, b] \subseteq [1, 2N_D]$ be an interval of integers with

$$\frac{1}{4} \cdot \frac{N}{D} < |L| < 2 \cdot \frac{N}{D} \quad \text{and} \quad a < \delta^{-C_d-3}.$$

Then at least one of the following alternatives holds:

1.

$$\# \left\{ 1 \leq c < D : \left| \sum_{n \in L} e(g(nc) - g(nD)) \right| > \delta \frac{N}{D} \right\} \ll_d \delta D.$$

2. There is a nonzero integer K such that

$$|K| \ll_d \delta^{-C_d}, \quad \|Kg\|_{C^\infty[N]} \ll_d \delta^{-C_d}.$$

This lemma, for fixed D , bounds the number of integers c with $1 \leq c < D$ for which the exponential sum over L exceeds $\delta N/D$. This is used to bound the number of bad base points noted in the proof strategy. The condition on L is taken for various adaptations. The condition on D is taken to ensure the argument of [2, §3] can be applied.

Proof. The only essential difference between this proof and the arguments in [2, §3] is the case when D is small. For the reader's convenience, we record the whole proof.

Apply Theorem 7 to the polynomials $\xi_c(n) = g(nc) - g(nD) = \sum_{j=1}^d \beta_j(c^j - D^j)n^j$ on L , at parameter $\delta/(4\pi)$ to absorb $\|e\|_{\text{Lip}}$. Write $\mathcal{R}$ for the exceptional set in alternative (1). Since [3] is stated for initial intervals starting at 1, translate $L = [a, b]$ to $[1, |L|]$ by $n \mapsto n + a - 1$. The shifted polynomial $\xi_c(n + a - 1)$ has coefficients differing from those of ξ_c by factors $a^{O(d)}$; the bound $a < \delta^{-C_d-3}$ keeps the $C^\infty[N_D]$ size $\delta^{-O_d(1)}$.

Let $C > 0$ be a parameter depending only on d .

Case $D \leq \delta^{-C}$. When D is small, we show that either each exponential sum is at most $\delta N/D$, or alternative (2) of the lemma holds. In other words, the exceptional set in alternative (1) of the lemma can be taken empty.

Applying Theorem 7 to each ξ_c , either the exponential sum is $\leq \delta N/D$, or the second alternative of that theorem produces q_c with $1 \leq |q_c| \ll_d \delta^{-O_d(1)}$ and

$$\|q_c \xi_c\|_{C^\infty[N_D]} \ll_d \delta^{-O_d(1)}.$$

Recall the definition of ξ_c , whose terms contain c and D . The second alternative of Theorem 7 therefore reads

$$N_D^j \|q_c \beta_j(c^j - D^j)\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)} \quad (1 \leq j \leq d).$$

To pass this to alternative (2) of the lemma, we need the property of g . So we put

$$K = q_c \prod_{i=1}^d (c^i - D^i), \quad K\beta_j = (q_c \beta_j(c^j - D^j)) \cdot \prod_{i \neq j} (c^i - D^i).$$

The second factor is an integer, so $\|K\beta_j\|_{\mathbb{R}/\mathbb{Z}}$ is controlled by $\|q_c \beta_j(c^j - D^j)\|_{\mathbb{R}/\mathbb{Z}}$. Changing scale from N_D to $N = N_D D$ costs only powers of D , and since $D \leq \delta^{-C}$ we have $|c^j - D^j| \leq D^j \leq \delta^{-Cj}$. Thus

$$0 < |K| \ll_d \delta^{-O_d(1)}, \quad N^j \|K\beta_j\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)},$$

which is alternative (2) of Lemma 10. Since for each c , either the exponential sum is at most $\delta N/D$, or alternative (2) of the lemma holds, we are done.

Case $D > \delta^{-C}$. If $|\mathcal{R}| \leq \delta D$ we are done. Otherwise each $c \in \mathcal{R}$ has $|\sum_{n \in L} e(\xi_c(n))| > \delta N/D$, so Theorem 7 supplies a nonzero integer q_c with $|q_c| \ll_d \delta^{-O_d(1)}$ and $\|q_c \xi_c\|_{C^\infty[N_D]} \ll_d \delta^{-O_d(1)}$. The

number of possible values of q_c is $\ll_d \delta^{-O_d(1)}$. Since $|\mathcal{R}| > \delta D > \delta^{-C+1}$, pigeonholing yields a single nonzero q and a set $E \subseteq \mathcal{R}$ with

$$|E| \gg_d \delta^{O_d(1)} D$$

such that $\|q\tilde{\xi}_c\|_{C^\infty[N_D]} \ll_d \delta^{-O_d(1)}$ for every $c \in E$. We choose $C > 0$ large enough depending only on d to make sure E is non-empty. In particular E has density $\gg_d \delta^{O_d(1)}$ in $\{1, \dots, D-1\}$, and

$$\|q\beta_j(c^j - D^j)\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)} (D/N)^j \quad (1 \leq j \leq d), \quad c \in E.$$

Fix $j \in \{1, \dots, d\}$ and set $\alpha = q\beta_j$. For every $c \in E$ we have

$$\|\alpha(c^j - D^j)\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)} \left(\frac{D}{N}\right)^j,$$

so αc^j lies in an arc of length $\varepsilon_0 \ll_d \delta^{-O_d(1)} (D/N)^j$ about αD^j .

To pass from the j th powers c^j to a denser set of integers, let $t = 2^j + 1$ and apply Lemma 9 to E , with density $\eta \gg_d \delta^{O_d(1)}$. This supplies $\gg_d \delta^{O_d(1)} D^j$ integers $\ell \in [tD^j]$ of the form $c_1^j + \dots + c_t^j$ with each $c_i \in E$.

For each such ℓ , the value $\alpha\ell$ lies in an arc I about $t\alpha D^j$ of length $\varepsilon \leq t\varepsilon_0 \ll_d \delta^{-O_d(1)} (D/N)^j$. The number of these ℓ is $\gg_d \delta^{O_d(1)} D^j$, a density $\sigma \gg_d \delta^{O_d(1)}$ in $\{1, \dots, tD^j\}$.

We apply Lemma 8: $\alpha\ell \in I$ for a density σ of the integers $\ell \leq tD^j$. From the bounds on ε and σ , one has $\varepsilon/\sigma \ll_d \delta^{-O_d(1)} (D/N)^j$. Thus $\varepsilon \leq \sigma/2$ once $(D/N)^j \ll_d \delta^{O_d(1)}$, which holds for $N/D > \delta^{-C_d}$ with C_d large in terms of d . The lemma produces an integer $r \ll_d \delta^{-O_d(1)}$ with

$$D^j \|r\alpha\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)} \left(\frac{D}{N}\right)^j,$$

hence $N^j \|rq\beta_j\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)}$. Doing the same for each j produces an integer r_j . Setting $K = q \prod_{j=1}^d r_j$ gives $0 < |K| \ll_d \delta^{-O_d(1)}$ and $\|Kg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}$, which is alternative (2) of Lemma 10. $\square$

5 Estimates on $\mathcal{F}_1$

As in the inclusion–exclusion decomposition, we can decompose the sum over the union of $\mathcal{F}_i$ into the sum over each of the $\mathcal{F}_i$ and the intersections. In this section, we deal with the summation over each $\mathcal{F}_i$. Without loss of generality, we can assume that $i = 1$, otherwise we can apply a permutation to change the coordinates. The contribution of $\mathcal{F}_1$ is given by the following proposition.

Proposition 11. *Let $s \geq 2$, and assume that $\mathcal{H}(s-1)$ holds. For every integer $d \geq 1$, there exist constants $C_d > 5$ and $c_0 = c_0(d, s) > 0$, with C_d depending only on d , such that the following holds. For every polynomial g of degree d , all N large enough depending on d and s , all δ with $N^{-c_0} < \delta < 1/8$, and all integers A in the window*

$$\delta^{-C_d} < A < \delta^{-(C_d+3)},$$

at least one of the following holds:

$$1. \quad |S_g(\mathcal{F}_1) - (s-1)! N^s| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

2. There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that $\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}$.

To prove the proposition, split

$$S_g(\mathcal{F}_1) = S_g(\mathcal{F}_1 \setminus \{n_1 = m_1\}) + S_g(\mathcal{F}_1 \cap \{n_1 = m_1\}) =: J_1 + J_2.$$

In the above, J_1 is the off-diagonal part, and J_2 is the diagonal part. The off-diagonal part J_1 has no main term. The diagonal part J_2 contains main term from $\mathcal{H}(s-1)$. In Section 5.1, we deal with the off-diagonal part J_1 , and in Section 5.2, we deal with the diagonal part J_2 .

Recall that the parameter A appears in the definition of $\mathcal{F}_1$, and is restricted to $\delta^{-C_d} < A < \delta^{-(C_d+3)}$. The lower bound matches the hypothesis $N/D > \delta^{-C_d}$ of Lemma 10. It is checked below, so the lemma can be applied.

5.1 Off-diagonal part J_1

The main result of this subsection is Lemma 14. Write $\mathcal{F}_1^\circ := \mathcal{F}_1 \setminus \{n_1 = m_1\}$ for the off-diagonal part of $\mathcal{F}_1$, so that $J_1 = S_g(\mathcal{F}_1^\circ)$.

To prove Lemma 14, we first need a parameterisation lemma for the off-diagonal part, which is slightly different from [10, Lemma 3.4] (the case $s = 2$) and the factorisations in [1, Lemma 2.3] and [11, §8].

Lemma 12 (Matrix parameterisation). *Let $H \geq 1$ and, with V as above, set*

$$V_H^\circ = \{(\vec{n}_s, \vec{m}_s) \in V : \gcd(n_1, m_1) \geq H, n_1 \neq m_1\}.$$

There is an injection $\phi: V_H^\circ \rightarrow M_{s \times s}(\mathbb{Z}_{>0})$ sending $(\vec{n}_s, \vec{m}_s)$ to a matrix

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1s} \\ a_{21} & a_{22} & \cdots & a_{2s} \\ \vdots & \vdots & \ddots & \vdots \\ a_{s1} & a_{s2} & \cdots & a_{ss} \end{pmatrix}$$

with $n_i = \prod_j a_{ij}$ the product of row i and $m_i = \prod_k a_{ki}$ the product of column i . Write π_1 for the projection onto the off-diagonal entries,

$$\pi_1((a_{ij})_{1 \leq i, j \leq s}) = (a_{ij})_{i \neq j},$$

forgetting the diagonal. For each $y = (a_{ij})_{i \neq j} \in \pi_1(\phi(V_H^\circ))$, writing

$$A_i = \prod_{j \neq i} a_{ij}, \quad B_i = \prod_{k \neq i} a_{ki}, \quad M_i = \max\{A_i, B_i\},$$

one has $A_1 \neq B_1$, and

$$\pi_1^{-1}(y) \cap \phi(V_H^\circ) = \{(a_{ij}) : (a_{ij})_{i \neq j} = y, \quad a_{11} \in [H, N/M_1], \quad a_{ii} \in [1, N/M_i] \ (2 \leq i \leq s)\}. \quad (2)$$

We take a moment to explain the intuition behind the parameterisation lemma. At $s = 2$, [10, Lemma 3.4] parameterises the solutions of $m_1 m_2 = n_1 n_2$ by

$$m_1 = ga, \quad m_2 = hb, \quad n_1 = gb, \quad n_2 = ha, \quad \gcd(a, b) = 1,$$

where $g = \gcd(m_1, n_1)$ and $h = \gcd(m_2, n_2)$.

To estimate the exponential sum over $\mathcal{F}_1^\circ$, it is important to find arithmetic progressions in the parameterisation. At $s = 2$, fix the coprime pair (a, b) . Both g and h stay free. Then we may first sum over g and h . As g runs, $n_1 = gb$ and $m_1 = ga$ are arithmetic progressions. As h runs, $n_2 = ha$ and $m_2 = hb$ are arithmetic progressions. This is the structure used in [10, Lemma 3.4] for the computation of the 4th moment. The lemma above keeps it and extends the parameterisation to general s . For general s , the lemma describes the fibres of π_1 on the image of ϕ . In particular, as a_{11} varies, $n_1 = a_{11}A_1$ and $m_1 = a_{11}B_1$ are arithmetic progressions. This is the most important property used in what follows, and the main addition compared with [1, Lemma 2.3] and [11, §8].

Proof. We first construct ϕ and check injectivity, then verify the fibre description.

Construction. Set $a_{ii} := \gcd(n_i, m_i)$ for every i . Write $u_i := n_i/a_{ii}$ and $v_i := m_i/a_{ii}$, so that $\gcd(u_i, v_i) = 1$ for each i and $\prod_i u_i = \prod_i v_i$. The off-diagonal entries a_{ij} with $i \neq j$ remain to be chosen so that

$$u_i = \prod_{j \neq i} a_{ij}, \quad v_i = \prod_{k \neq i} a_{ki}.$$

For the off-diagonal entries, the way we choose them does not matter so much as long as the procedure makes the parameterisation unique. One may instead fill them by [1, Lemma 2.3] or by the construction of [11, §8]. For the reader's convenience we record the following construction. We proceed by induction on s . Extract the off-diagonal entries of the first row by successive gcds

$$a_{1j} = \gcd\left(u_1 / \prod_{2 \leq t < j} a_{1t}, v_j\right) \quad (j \geq 2).$$

We claim that $u_1 = \prod_{j=2}^s a_{1j}$. In fact, from $\prod_i u_i = \prod_i v_i$ we have $u_1 \mid \prod_i v_i$. Since also $\gcd(u_1, v_1) = 1$, we get $u_1 \mid \prod_{j=2}^s v_j$. Write L for the leftover $u_1 / \prod_{j=2}^s a_{1j}$. Then L divides $\prod_{j=2}^s v_j$, but is coprime to each v_j with $j \geq 2$, so $L = 1$. Similarly, extract the off-diagonal entries of the first column by successive gcds

$$a_{i1} = \gcd\left(u_i, v_1 / \prod_{2 \leq t < i} a_{t1}\right) \quad (i \geq 2).$$

The same argument gives $v_1 = \prod_{i=2}^s a_{i1}$.

When $s = 2$, the off-diagonal consists exactly of a_{12} and a_{21} , so $u_1 = a_{12}$ and $v_1 = a_{21}$. The identity $u_1 u_2 = v_1 v_2$ together with $\gcd(u_2, v_2) = 1$ then gives $u_2 = a_{21}$ and $v_2 = a_{12}$.

Now suppose $s \geq 3$. Write $u'_i := u_i/a_{i1}$ and $v'_i := v_i/a_{1i}$ for $i = 2, \dots, s$. These are positive integers with $\gcd(u'_i, v'_i) = 1$. Cancelling $u_1 = \prod_{j \geq 2} a_{1j}$ and $v_1 = \prod_{i \geq 2} a_{i1}$ from the product equation gives

$$u'_2 \cdots u'_s = v'_2 \cdots v'_s.$$

Thus the same construction applies to u'_i, v'_i , and uniquely determines the remaining $(s-1) \times (s-1)$ block. That is,

$$u'_i = \prod_{\substack{j=2 \\ j \neq i}}^s a_{ij}, \quad v'_i = \prod_{\substack{k=2 \\ k \neq i}}^s a_{ki} \quad (i = 2, \dots, s).$$

and therefore

$$u_i = a_{i1} u'_i = \prod_{j \neq i} a_{ij}, \quad v_i = a_{1i} v'_i = \prod_{k \neq i} a_{ki}.$$

Together with $a_{ii} = \gcd(n_i, m_i)$, this gives $n_i = \prod_j a_{ij}$ and $m_i = \prod_k a_{ki}$. Note that in this construction, $\gcd(A_i, B_i) = 1$ for $i = 1, \dots, s$. Also note that this construction procedure makes the parameterisation unique.

Injectivity. The map is injective because the pair is recovered as the row and column products $n_i = \prod_j a_{ij}$ and $m_i = \prod_k a_{ki}$.

Fibre. Fix $y = (a_{ij})_{i \neq j} \in \pi_1(\phi(V_H^\circ))$, and write A_i, B_i and M_i as in the statement. Since y arises from a matrix in $\phi(V_H^\circ)$, one has $A_1 \neq B_1$ and $\gcd(A_1, B_1) = 1$ by the construction.

We check the two inclusions in (2). We first show that $\pi_1^{-1}(y) \cap \phi(V_H^\circ)$ is contained in the right-hand side. If $(a_{ij}) \in \pi_1^{-1}(y) \cap \phi(V_H^\circ)$, then $(a_{ij}) = \phi(\vec{n}_s, \vec{m}_s)$ for some $(\vec{n}_s, \vec{m}_s) \in V_H^\circ$, and

$$n_i = a_{ii}A_i, \quad m_i = a_{ii}B_i \quad (1 \leq i \leq s).$$

Membership in V_H° and $\gcd(A_1, B_1) = 1$ give $a_{11} = \gcd(n_1, m_1) \geq H$. The bounds $n_i, m_i \leq N$ give $a_{ii} \leq N/M_i$ for each i . Thus (a_{ij}) lies in the right-hand side of (2).

We next show that the right-hand side of (2) is contained in $\pi_1^{-1}(y) \cap \phi(V_H^\circ)$. Let $a_{11} \in [H, N/M_1]$ and $a_{ii} \in [1, N/M_i]$ for $2 \leq i \leq s$, and let (a_{ij}) be the matrix with this diagonal and with off-diagonal entries given by y . Reconstruct

$$n_i = a_{ii}A_i, \quad m_i = a_{ii}B_i \quad (1 \leq i \leq s).$$

We claim that $(\vec{n}_s, \vec{m}_s)$ lies in V_H° and that $\phi(\vec{n}_s, \vec{m}_s) = (a_{ij})$.

We first check membership in V . The bounds $a_{ii} \leq N/M_i$ and $A_i, B_i \leq M_i$ give $n_i, m_i \leq N$. The coordinates are positive integers. The product identity $n_1 \cdots n_s = m_1 \cdots m_s$ holds because $\prod_i A_i$ and $\prod_i B_i$ are both the product of the off-diagonal entries of y .

We next check the gcd condition. Since $\gcd(A_1, B_1) = 1$, we have $\gcd(n_1, m_1) = a_{11} \geq H$. Finally, $A_1 \neq B_1$ and $a_{11} \geq 1$ give $n_1 = a_{11}A_1 \neq a_{11}B_1 = m_1$. Thus the reconstructed tuple lies in V_H° .

It remains to check that $\phi(\vec{n}_s, \vec{m}_s) = (a_{ij})$. Since $\gcd(A_i, B_i) = 1$ for each i , the diagonal of $\phi(\vec{n}_s, \vec{m}_s)$ is recovered as $a_{ii} = \gcd(n_i, m_i)$. Uniqueness of the parameterisation then forces the off-diagonal to be exactly y . $\square$

Remark 13. The same parameterisation ϕ applies to any finite set of positive solutions of $n_1 \cdots n_s = m_1 \cdots m_s$ (in particular to the scaled images in Piece III of Lemma 19).

We now deal with J_1 . The following lemma is the goal of this section.

Lemma 14. Let the parameters be as in Proposition 11. At least one of the following holds:

1. $|J_1| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}$.
2. There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that $\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}$.

We now explain the proof strategy for Lemma 14.

Apply Lemma 12 with $H = A$ to $\mathcal{F}_1^\circ = V_A^\circ$, and set

$$\mathcal{F}'_1 := \pi_1(\phi(\mathcal{F}_1^\circ))$$

for the resulting set of off-diagonal base points. In Figure 1, ϕ is the parameterisation and π_1 is the projection; π_1 forgets the diagonal. Each point of $\mathcal{F}'_1$ is a base point. The fibre over it is given by (2). Figure 2 draws each fibre as a vertical interval.

A first attempt uses only Lemma 12 and $|e| = 1$. Each fibre has length at most $\prod_{i=1}^s N/M_i$, and summing over the base points in $\mathcal{F}'_1$ gives

$$|J_1| \ll N^s (\log N)^{s(s-1)} \ll_s N^s (\log N)^{O(s^2)}.$$

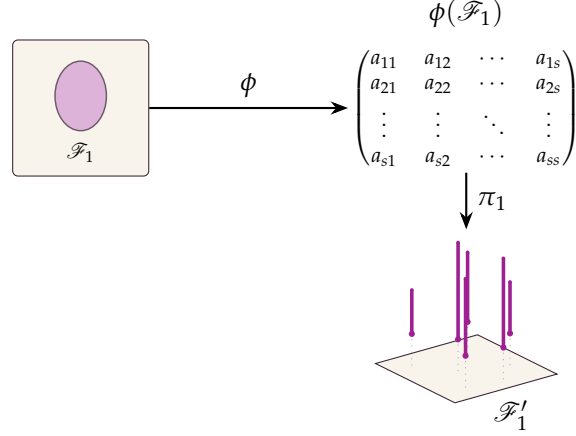


Figure 2: Each point of $\mathcal{F}'_1$ is a base point. The vertical intervals are the fibres of (2).

This is even larger than the size of the main term $s! N^s$, so this is not enough.

The proof uses Lemma 10 to get a δ saving. We express J_1 by summing first over the fibres with fixed base point, then summing over all base points. The base points are then split into good and bad according to the size of the fibre sum. For the bad base points, we show that they take up only a small proportion by using Lemma 10.

Proof. Assume alternative (2) of the present lemma fails. We start from $J_1 = S_g(\mathcal{F}_1^\circ)$. Lemma 12 with $H = A$ sends each $(\vec{n}_s, \vec{m}_s) \in \mathcal{F}_1^\circ$ to a matrix (a_{ij}) with $n_i = \prod_j a_{ij}$ and $m_i = \prod_k a_{ki}$. Recall that

$$\mathcal{F}'_1 = \pi_1(\phi(\mathcal{F}_1^\circ)).$$

For each $(a_{ij})_{i \neq j} \in \mathcal{F}'_1$, Lemma 12 gives the fibre (2) with $H = A$, and $n_i = a_{ii}A_i$, $m_i = a_{ii}B_i$ for $1 \leq i \leq s$. Substituting into J_1 gives

$$J_1 = \sum_{\mathcal{F}'_1} \sum_{\substack{a_{11} \in [A, N/M_1] \\ a_{ii} \in [1, N/M_i] \ (i \geq 2)}} \prod_{i=1}^s e(g(a_{ii}A_i) - g(a_{ii}B_i)).$$

The outer sum runs over the base points, and the inner sum is the exponential sum over one fibre, as in Figure 2. Since $|e| = 1$, the sums over a_{ii} for $i \geq 2$ are at most N/M_i , and we obtain

$$|J_1| \leq \sum_{\mathcal{F}'_1} \left| \sum_{A \leq a_{11} \leq N/M_1} e(g(a_{11}A_1) - g(a_{11}B_1)) \right| \prod_{i=2}^s \frac{N}{M_i}. \quad (3)$$

By symmetry between A_1 and B_1 , sum separately on $\{A_1 < B_1\}$ and $\{A_1 > B_1\}$; we treat $\{A_1 < B_1\}$ (so $M_1 = B_1$) and the other side is identical.

For a base point $(a_{ij})_{i \neq j} \in \mathcal{F}'_1$, write

$$S = \sum_{A \leq a_{11} \leq N/B_1} e(g(a_{11}A_1) - g(a_{11}B_1)).$$

We call the right endpoint N/B_1 the height of the fibre. The lower bound $A > \delta^{-C_d}$ is imposed so that this height exceeds δ^{-C_d} , as required by Lemma 10. This is checked in the proof of the following claim. The strategy is to apply Lemma 10 to S and split the fibres of Figure 2 into good and bad. On a good fibre, $|S| \ll \delta N/B_1$. The bad fibres are those for which the exponential sum is large. This is recorded as the following claim.

Claim 1. Let $(a_{ij})_{i \neq j} \in \mathcal{F}'_1$ be a base point with $A_1 < B_1$. For S as above, there exists a set $\mathcal{R}_{B_1} \subset \{1, \dots, B_1 - 1\}$, depending only on B_1 , with $|\mathcal{R}_{B_1}| \ll_d \delta B_1$ such that

$$|S| \leq 2\delta \frac{N}{B_1} + \mathbf{1}_{A_1 \in \mathcal{R}_{B_1}} \cdot \frac{N}{B_1}.$$

Proof of the claim. We plan to apply Lemma 10. But the input of this lemma requires the length of the interval to be comparable to the height. Since there are long fibres and short fibres as shown in Figure 3, one may worry that if the fibre is too short, then we are not able to apply Lemma 10.

But the observation is that we can still apply Lemma 10, because the height of the fibre is large enough. We divide into two cases.

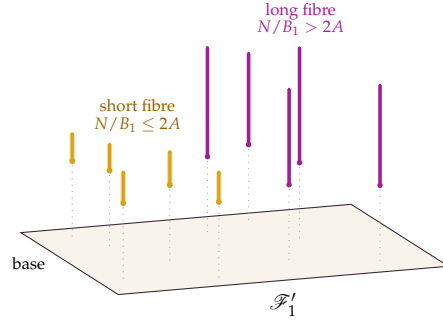


Figure 3: The fibre interval $L = [A, N/B_1]$ is drawn floating above $\mathcal{F}'_1$. Gold: short fibres $A \leq N/B_1 \leq 2A$. Purple: long fibres $N/B_1 > 2A$.

Long fibres ($N/B_1 > 2A$). In this case the range $L := [A, N/B_1]$ is an interval of length $\asymp N/B_1$ with left endpoint $A < \delta^{-(C_d+3)}$, so alternative (1) of Lemma 10 applies directly. There is $\mathcal{R}_{B_1} \subset \{1, \dots, B_1 - 1\}$ with $|\mathcal{R}_{B_1}| \ll_d \delta B_1$ such that $|S| \leq \delta N/B_1$ for all $A_1 \notin \mathcal{R}_{B_1}$. Without the length comparison, L would be an illegal input for that lemma.

Short fibres. Here $A \leq N/B_1 \leq 2A$, so $N/B_1 \asymp A$. We write S as a difference of two initial segments,

$$S = \sum_{1 \leq a_{11} \leq N/B_1} e(g(a_{11}A_1) - g(a_{11}B_1)) - \sum_{1 \leq a_{11} \leq A-1} e(g(a_{11}A_1) - g(a_{11}B_1)).$$

Both $[1, N/B_1]$ and $[1, A-1]$ then have length comparable to N/B_1 , and both start at 1, so Lemma 10 applies to each. Let $\mathcal{R}_{B_1}$ be the union of the two exceptional sets. One still has $|\mathcal{R}_{B_1}| \ll_d \delta B_1$. For $A_1 \notin \mathcal{R}_{B_1}$, each of the two sums is at most $\delta N/B_1$, and therefore $|S| \leq 2\delta N/B_1$. The factor 2 is the only extra cost of writing the short fibre as a difference of two longer sums.

In either case one has $|S| \leq 2\delta N/B_1$ off a set $\mathcal{R}_{B_1}$ of size $\ll_d \delta B_1$. On that set the trivial bound $|S| \leq N/B_1$ gives the claimed inequality.

This proves the claim.

Recall that S is obtained by fixing the base point $(a_{ij})_{i \neq j}$ and summing over a_{11} . The claim therefore groups the base points, or equivalently the fibres, into good and bad: the good fibres contribute the first term, with a 2δ saving, while the bad fibres—those $(a_{ij})_{i \neq j}$ with $A_1 \in \mathcal{R}_{B_1}$ —contribute the second term. By the claim one has $|\mathcal{R}_{B_1}| \ll_d \delta B_1$, so the number of bad fibres is small, which will be helpful when summing over the base points.

Next we sum over the base points, that is, $\mathcal{F}'_1$. Since $M_1 = B_1$ in this branch, $\frac{N}{B_1} \prod_{i=2}^s \frac{N}{M_i} = \prod_{i=1}^s \frac{N}{M_i}$. Write

$$\Sigma_I := \sum_{\mathcal{F}'_1 \cap \{A_1 < B_1\}} \prod_{i=1}^s \frac{N}{M_i}, \quad \Sigma_{II} := \sum_{\mathcal{F}'_1 \cap \{A_1 < B_1, A_1 \in \mathcal{R}_{B_1}\}} \prod_{i=1}^s \frac{N}{M_i}.$$

Inserting the claim, the contribution of $\{A_1 < B_1\}$ to the right-hand side of (3) satisfies

$$\sum_{\mathcal{F}'_1 \cap \{A_1 < B_1\}} |S| \prod_{i=2}^s \frac{N}{M_i} \leq 2\delta \Sigma_I + \Sigma_{II}.$$

The term Σ_I . Since $M_i \geq A_i := \prod_{j \neq i} a_{ij}$ for every $i = 1, \dots, s$, we may replace each M_i by A_i and drop the restriction $A_1 < B_1$. Summing row by row,

$$\Sigma_I \leq \sum_{\mathcal{F}'_1} \prod_{i=1}^s \frac{N}{A_i} \leq \prod_{i=1}^s \left(N \sum_{A_i \leq N} \frac{\tau_{s-1}(A_i)}{A_i} \right).$$

By Lemma 4, $\sum_{A_i \leq N} \tau_{s-1}(A_i)/A_i \leq (2 \log N)^{s-1}$ for each of the s rows, so

$$\Sigma_I \ll N^s (\log N)^{s(s-1)} \ll_s N^s (\log N)^{O(s^2)}.$$

The term Σ_{II} . We are still in the branch $A_1 < B_1$, so $M_1 = B_1$. For $i \geq 2$, we have $M_i \geq A_i$, and we use this trivial bound $\prod_{i=2}^s N/M_i \leq \prod_{i=2}^s N/A_i$. So Σ_{II} is less than

$$\sum_{\mathcal{F}'_1 \cap \{A_1 < B_1, A_1 \in \mathcal{R}_{B_1}\}} \frac{N}{B_1} \prod_{i=2}^s \frac{N}{A_i}. \quad (4)$$

The following picture, at $s = 5$, records which entries of the matrix in Lemma 12 appear in this denominator; the A_i and B_i are the off-diagonal row and column products.

A_1	$\cdot$	a_{12}	a_{13}	a_{14}	a_{1s}
A_2	a_{21}	$\cdot$	a_{23}	a_{24}	a_{2s}
A_3	a_{31}	a_{32}	$\cdot$	a_{34}	a_{3s}
A_4	a_{41}	a_{42}	a_{43}	$\cdot$	a_{4s}
A_s	a_{s1}	a_{s2}	a_{s3}	a_{s4}	$\cdot$
	B_1	B_2	B_3	B_4	B_s

The first row is white: those entries do not appear in the denominator. The darker cells appear twice in the denominator, and the paler cells appear once. Since the first row does not appear in the denominator, we may first fix rows $2, \dots, s$ and sum over the first row, without worrying about the weight changing in (4).

Fix rows $2, \dots, s$. Then B_1 and $A_2, \dots, A_s$ are constant, so the weight in (4) stays constant, and we sum first over the first row. For each fixed A_1 , the number of tuples $(a_{12}, \dots, a_{1s})$ with $A_1 = \prod_{j=2}^s a_{1j}$ is at most $\tau_{s-1}(A_1)$. Thus the inner sum over the first row contributes a factor at most

$$\sum_{\substack{A_1 < B_1 \\ A_1 \in \mathcal{R}_{B_1}}} \tau_{s-1}(A_1) \leq \sum_{A_1 \in \mathcal{R}_{B_1}} \tau_{s-1}(A_1).$$

By Cauchy–Schwarz and the second bound of Lemma 4, we have

$$\begin{aligned} \sum_{A_1 \in \mathcal{R}_{B_1}} \tau_{s-1}(A_1) &\leq |\mathcal{R}_{B_1}|^{1/2} \left(\sum_{n \leq B_1} \tau_{s-1}(n)^2 \right)^{1/2} \\ &\ll_{d,s} \delta^{1/2} B_1 (\log N)^{O(s^2)}. \end{aligned}$$

We next sum over rows $2, \dots, s$. Multiplying the above count by $\frac{N}{B_1} \prod_{i=2}^s \frac{N}{A_i}$ as in (4) and summing over rows $2, \dots, s$, the factors of B_1 cancel and we obtain

$$\Sigma_{\text{II}} \ll_{d,s} \delta^{1/2} N (\log N)^{O(s^2)} \sum_{\text{rows } 2, \dots, s} \prod_{i=2}^s \frac{N}{A_i}.$$

The remaining sum is a product of row-sums exactly as for Σ_{I} , so

$$\Sigma_{\text{II}} \ll_d \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

Combining, we have $2\delta\Sigma_{\text{I}} + \Sigma_{\text{II}} \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}$. The same bound holds on $\{A_1 > B_1\}$ by symmetry. Thus, unless the Diophantine alternative of Lemma 10 occurs, $|J_1| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}$. If it does occur, the integer K produced there is the q of alternative (2). $\square$

5.2 Diagonal part J_2

We now deal with J_2 . The following lemma is the goal of this section. The argument uses the inductive hypothesis $\mathcal{H}(s-1)$.

Lemma 15. *Assume that $\mathcal{H}(s-1)$ holds. There exists $c_0 = c_0(d, s) > 0$ such that, for the remaining parameters as in Proposition 11, with this choice of c_0 , at least one of the following alternatives occurs:*

1. $|J_2 - (s-1)! N^s| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}$.
2. There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that $\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}$.

Proof. On $\mathcal{F}_1 \cap \{n_1 = m_1\}$ the constraint $\gcd(n_1, m_1) \geq A$ defining $\mathcal{F}_1$ reduces to $n_1 \geq A$, and the diagonal factor $e(g(n_1))\overline{e(g(n_1))}$ equals 1. Summing freely over $n_1 \in [A, N]$ and over the remaining variables, which satisfy $n_2 \cdots n_s = m_2 \cdots m_s$,

$$J_2 = (N - A + 1) U_{s-1}(N).$$

Writing $U_{s-1}(N) = N^{s-1} \mathcal{M}_{s-1}(N)$ and expanding $(N - A + 1)N^{s-1} = N^s + O(AN^{s-1})$,

$$J_2 = N^s \mathcal{M}_{s-1}(N) + O(AN^{s-1} \mathcal{M}_{s-1}(N)). \quad (5)$$

If $s = 2$, $\mathcal{H}(1)$ gives $\mathcal{M}_1(N) = 1$ exactly, so (5) reads $J_2 = N^2 + O(AN)$. The upper bound $A < \delta^{-(C_d+3)}$ together with $\delta > N^{-c_0}$ (with c_0 small enough) yields $A \ll_d \delta N$, hence $O(AN) \ll_d \delta N^2$, which is alternative (1).

If $s \geq 3$, apply the inductive hypothesis $\mathcal{H}(s-1)$ to $\mathcal{M}_{s-1}(N)$. If it produces alternative (2), the same q serves here and we are done. Otherwise

$$|\mathcal{M}_{s-1}(N) - (s-1)!| \ll_{d,s} \delta (\log N)^{O(s^2)},$$

and so $\mathcal{M}_{s-1}(N) \ll_{d,s} (\log N)^{O(s^2)}$. The same $A \ll_d \delta N$ also bounds the error term. Hence (5) gives

$$|J_2 - (s-1)! N^s| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)},$$

which is alternative (1). $\square$

Combining Lemmas 14 and 15 proves Proposition 11.

Remark 16 (Symmetry among $\mathcal{F}_i$). For $i \in \{1, \dots, s\}$, relabel $(n_1, m_i) \mapsto (m_i, n_1)$ and permute the remaining indices so that $\mathcal{F}_i$ becomes $\mathcal{F}_1$ with $\gcd(n_1, m_1) \geq A$ unchanged. Proposition 11 therefore applies identically to $\sum_{\mathcal{F}_i}$, with the same main term $(s-1)! N^s$ and the same error bound.

6 Estimates on intersections

For $2 \leq \kappa \leq s$ and distinct indices $\{i_1, \dots, i_\kappa\} \subset \{1, \dots, s\}$, we bound sums over $\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa}$. These intersections contribute no N^s main term.

Proposition 17 (General intersection). Let $C_d > 5$ be as in Proposition 11. Let $s \geq 2$ and let $2 \leq \kappa \leq s$. Let $\{i_1, \dots, i_\kappa\} \subset \{1, \dots, s\}$ be distinct. For every polynomial g of degree d , all N large enough depending on d and s , all δ with $0 < \delta < 1/8$, and all integers A in the window

$$\delta^{-(C_d+2)} < A < \delta^{-(C_d+3)},$$

at least one of the following holds:

1.

$$|S_g(\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa})| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

2. There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that $\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}$.

Without loss of generality, we may assume that $i_1 = 1$, since otherwise we may relabel the m -variables. We then split

$$\begin{aligned} S_g(\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa}) &= S_g(\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa} \cap \{n_1 = m_1\}) \\ &\quad + S_g(\mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa} \setminus \{n_1 = m_1\}) \\ &=: I_2 + I_1, \end{aligned}$$

as in the $J_1 + J_2$ split of Proposition 11. In Section 6.1 we deal with the off-diagonal part I_1 , and in Section 6.2 we deal with the diagonal part I_2 .

6.1 Off-diagonal part I_1

We now deal with I_1 . Write

$$\mathcal{F}_I^\circ := \mathcal{F}_{i_1} \cap \dots \cap \mathcal{F}_{i_\kappa} \setminus \{n_1 = m_1\}$$

for the off-diagonal part of the intersection (with $i_1 = 1$).

Similar to the J_1 case, we first need a parameterisation lemma to extract the structure of arithmetic progressions. But different from the J_1 case, the multiple intersections of $\mathcal{F}_I^\circ$ have more constraints than a single $\mathcal{F}_1$, making it hard to directly apply the matrix parameterisation of

Lemma 12. The difficulty coming from the multiple constraints is that the extra large gcd still involves the same n_1 , and therefore cuts the fibre in a_{11} .

For example, take $i_1 = 1$ and $i_2 = 2$, that is $\mathcal{F}_1 \cap \mathcal{F}_2$. On $\mathcal{F}_1$, Lemma 12 writes $n_1 = a_{11}A_1$ and $m_1 = a_{11}B_1$, with fibre $a_{11} \in [A, N/M_1]$. Membership in $\mathcal{F}_2$ is $\gcd(n_1, m_2) \geq A$. Equivalently, $\gcd(a_{11}A_1, m_2) \geq A$. The admissible a_{11} must share a large factor with m_2 , so the fibre is no longer an interval, and Lemma 10 does not apply.

To make this strategy work, we introduce new parameters $t_2, \dots, t_s$ to separate the large gcds from the other factors. Hence the remaining parameters are fully characterised by relatively coprime conditions, which makes it convenient to pass to exponential sums over arithmetic progressions. More precisely, we have the following lemma:

Lemma 18 (Intersection parameterisation). *Let $I = \{i_1, \dots, i_\kappa\} \subseteq \{1, \dots, s\}$ with $i_1 = 1$. There is an injection $\psi: \mathcal{F}_I^\circ \rightarrow \mathbb{Z}_{>0}^{3s}$ sending $(n_1, \dots, n_s, m_1, \dots, m_s)$ to (b, v) , where*

$$v := (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s).$$

Denote $T := t_2 \cdots t_s$. The pair (b, v) satisfies

$$\begin{pmatrix} n_1 & n_2 & \cdots & n_s \\ m_1 & m_2 & \cdots & m_s \end{pmatrix} = \begin{pmatrix} bTn'_1 & n_2 & \cdots & n_s \\ bTm'_1 & t_2m'_2 & \cdots & t_sm'_s \end{pmatrix},$$

together with

$$\gcd(n'_1, m'_1) = 1, \quad n'_1 \neq m'_1.$$

Write π_2 for the projection onto the outer parameters,

$$\pi_2(b, v) = v,$$

forgetting b . For each $v \in \pi_2(\psi(\mathcal{F}_I^\circ))$, we have

$$\pi_2^{-1}(v) \cap \psi(\mathcal{F}_I^\circ) = \mathcal{G}(v),$$

where

$$\mathcal{G}(v) = \left\{ (b, v) : \frac{A}{T} \leq b \leq \frac{N}{T \max\{n'_1, m'_1\}}, \quad \gcd(b, m'_j) = 1 \text{ for } 2 \leq j \leq s \right\}.$$

We take a moment to explain this lemma. As in the $\mathcal{F}_1$ case we want to have a parameterisation and then conduct exponential sums over the fibres, where the base points can be grouped into good and bad parts. The outer parameters v are the new base points, playing the role of $\mathcal{F}'_1$. The fibre variable b plays the role of a_{11} . The factors t_j are introduced to restore the information of the common divisors. The lemma describes the fibres of π_2 on the image of ψ . The fibre is exactly one interval satisfying some coprime conditions, where the coprime conditions can be removed by Möbius inversion in the following arguments.

Proof. We first construct ψ and check injectivity, then verify the fibre description.

Construction. Set

$$b_1 = \gcd(n_1, m_1), \quad n'_1 = \frac{n_1}{b_1}, \quad m'_1 = \frac{m_1}{b_1}.$$

Then $\gcd(n'_1, m'_1) = 1$. Recursively, for $j = 2, \dots, s$, define

$$t_j = \gcd(b_{j-1}, m_j), \quad b_j = \frac{b_{j-1}}{t_j}, \quad m'_j = \frac{m_j}{t_j}.$$

Write b for b_s , and write T for $t_2 \cdots t_s$. Thus $b_1 = bT$, so $n_1 = bTn'_1$ and $m_1 = bTm'_1$. Note that from the construction, one immediately sees that $n'_1 \neq m'_1$, that $\gcd(b_j, m'_j) = 1$ for $2 \leq j \leq s$, and hence that $\gcd(b, m'_j) = 1$. Also note that this procedure uniquely determines the parameterisation.

Injectivity. The map is injective because the pair is recovered as $n_1 = bTn'_1$, $m_1 = bTm'_1$, the coordinates n_i for $i \geq 2$ as in ν , and $m_j = t_j m'_j$ for $j \geq 2$.

Fibre. Fix a $\nu \in \pi_2(\psi(\mathcal{F}_I^\circ))$. Write

$$\nu = (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s),$$

and denote $T := t_2 \cdots t_s$. Note that this ν arises from a construction. So it naturally satisfies the properties $\gcd(n'_1, m'_1) = 1$ and $n'_1 \neq m'_1$ as noted in the construction. We check the two inclusions between the fibre $\pi_2^{-1}(\nu) \cap \psi(\mathcal{F}_I^\circ)$ and $\mathcal{G}(\nu)$.

We first show that $\pi_2^{-1}(\nu) \cap \psi(\mathcal{F}_I^\circ)$ is contained in $\mathcal{G}(\nu)$. Take an element of this fibre: it is of the form $(b, \nu) = \psi(n_1, \dots, n_s, m_1, \dots, m_s)$ for some $(n_1, \dots, n_s, m_1, \dots, m_s) \in \mathcal{F}_I^\circ$. We first check the range of b . Since $n_1 = bTn'_1$ and $m_1 = bTm'_1$, the bounds $n_1, m_1 \leq N$ give

$$b \leq \frac{N}{T \max\{n'_1, m'_1\}}.$$

The point lies in $\mathcal{F}_1$, so $\gcd(n_1, m_1) \geq A$. From the construction this gcd equals bT , hence $b \geq A/T$. The coprime conditions $\gcd(b, m'_j) = 1$ for $2 \leq j \leq s$ likewise come from the construction. Thus $(b, \nu) \in \mathcal{G}(\nu)$.

We next show that $\mathcal{G}(\nu)$ is contained in $\pi_2^{-1}(\nu) \cap \psi(\mathcal{F}_I^\circ)$. Let (b, ν) be a point of $\mathcal{G}(\nu)$. Our strategy is to reconstruct the integral vector $(n_1, \dots, n_s, m_1, \dots, m_s)$ from (b, ν) and check that it lies in $\mathcal{F}_I^\circ$ and that its image under ψ is exactly (b, ν) .

We reconstruct $n_1 = bTn'_1$ and $m_1 = bTm'_1$, together with n_i as in ν for $i \geq 2$ and $m_j = t_j m'_j$ for $j \geq 2$.

We first check that the reconstructed integral vector $(n_1, \dots, n_s, m_1, \dots, m_s)$ lies in $\mathcal{F}_I^\circ$. In order to check this, we must check that it lies in V , that it lies in every $\mathcal{F}_i$ for $i \in I$, and that $n_1 \neq m_1$. Note that $(b, \nu) \in \mathcal{G}(\nu)$, so b satisfies

$$\frac{A}{T} \leq b \leq \frac{N}{T \max\{n'_1, m'_1\}}, \quad \gcd(b, m'_j) = 1 \text{ for } 2 \leq j \leq s.$$

We first check membership in V . The upper bound $b \leq N/(T \max\{n'_1, m'_1\})$ gives $n_1 = bTn'_1 \leq N$ and $m_1 = bTm'_1 \leq N$. The coordinates $n_2, \dots, n_s$ appear in ν , hence are at most N . For $j \geq 2$, the reconstructed $m_j = t_j m'_j$ is the corresponding coordinate of the original point in V that produced ν , hence is also at most N . The coordinates are positive integers. The product identity holds because ν arises from a point in the image of ψ , so $n'_1 n_2 \cdots n_s = m'_1 (t_2 m'_2) \cdots (t_s m'_s)$, and multiplying by bT gives the identity $n_1 \cdots n_s = m_1 \cdots m_s$.

We next check membership in $\mathcal{F}_1$. Note that $\gcd(n'_1, m'_1) = 1$, so $\gcd(n_1, m_1) = bT \geq A$. We next check the condition $n_1 \neq m_1$. As recorded above one has $n'_1 \neq m'_1$, and multiplying by $bT \geq 1$ yields $n_1 = bTn'_1 \neq bTm'_1 = m_1$.

We next check membership in each $\mathcal{F}_i$ for $i \in I \setminus \{1\}$. For each such i , reconstruction gives $\gcd(n_1, m_i) = \gcd(bTn'_1, t_i m'_i)$. Since $t_i \mid T$, we may factor t_i out of the gcd:

$$\gcd(bTn'_1, t_i m'_i) = t_i \gcd(b(T/t_i)n'_1, m'_i).$$

Since $\gcd(b, m'_i) = 1$, the remaining gcd drops b :

$$\gcd(b(T/t_i)n'_1, m'_i) = \gcd((T/t_i)n'_1, m'_i).$$

Hence $\gcd(n_1, m_i) = \gcd(Tn'_1, t_i m'_i)$, which is independent of b . Thus as b varies in $\mathcal{G}(v)$, the reconstructed $(n_1, \dots, n_s, m_1, \dots, m_s)$ has constant gcd. Since $v \in \pi_2(\psi(\mathcal{F}_I^\circ))$, there exists an element $\xi \in \mathcal{F}_I^\circ$ such that $v = \pi_2(\psi(\xi))$. We may write $\psi(\xi) = (b_0, v)$. By the first inclusion, (b_0, v) is contained in $\mathcal{G}(v)$. Since we have shown above that the reconstructed tuple has constant gcd throughout the fibre $\mathcal{G}(v)$, we have that the gcds $\gcd(n_1, m_i)$ for $i \in I \setminus \{1\}$ are all the same as those of the original element ξ . Thus the reconstructed tuple lies in $\mathcal{F}_I^\circ$.

Next we check that the image of the reconstructed integral vector $(n_1, \dots, n_s, m_1, \dots, m_s)$ under ψ is exactly (b, v) . Since v arises from the construction, one has $\gcd(n'_1, m'_1) = 1$, so the first step recovers $b_1 = bT$. Since the procedure uniquely determines the parameterisation, and $\gcd(b, m'_j) = 1$ for $2 \leq j \leq s$, the successive steps recover the same t_j and m'_j , and thus $\psi(n_1, \dots, n_s, m_1, \dots, m_s) = (b, v)$. $\square$

We now deal with I_1 . The following lemma is the goal of this section.

Lemma 19 (Off-diagonal part I_1). *Let the parameters be as in Proposition 17. Recall*

$$I_1 = S_g(\mathcal{F}_I^\circ).$$

At least one of the following holds:

1. $|I_1| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$
2. *There exists $q \in \mathbb{Z} \setminus \{0\}$ with $|q| \ll_d \delta^{-O_d(1)}$ such that $\|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}.$*

We now explain the proof strategy for Lemma 19. Write $\mathcal{P} = \pi_2(\psi(\mathcal{F}_I^\circ))$ for the finite set of outer parameters. The strategy is illustrated in Figure 4.

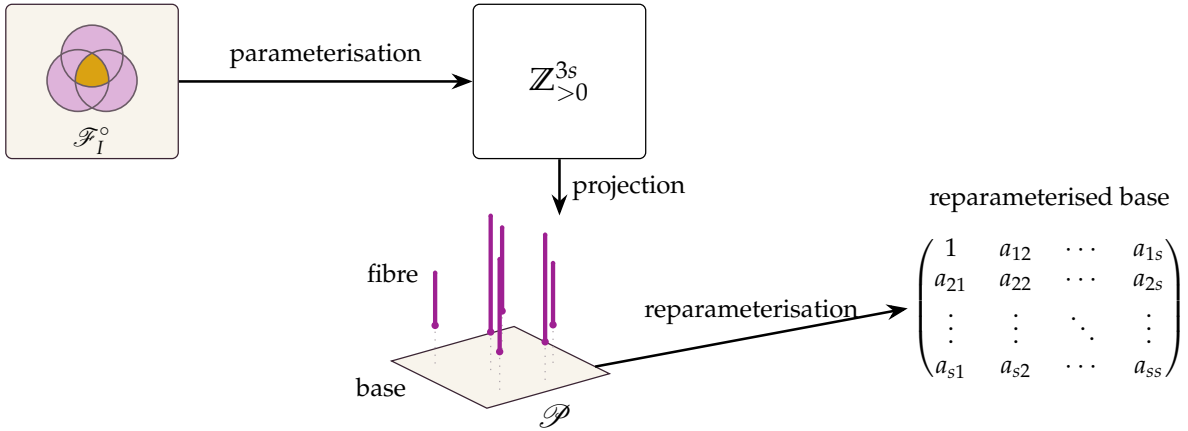


Figure 4: The map ψ of Lemma 18 sends $\mathcal{F}_I^\circ$ into $\mathbb{Z}_{>0}^{3s}$. The projection π_2 forgets the first coordinate leaving the base $\mathcal{P}$ with a fibre. We group the fibres in the same spirit as in Lemma 14. A reparameterisation of the base is needed.

The estimate follows Figure 4 in four steps. From parameterisation Lemma 18, we may bound I_1 by first summing over the fibre and then summing over the base. The idea is similar to the proof of Lemma 14, but we have to overcome more difficulties. The first difficulty is that the fibre is not an interval, but a more complicated set with coprime constraints. By using Möbius inversion, we may remove the coprime constraints by introducing a new parameter ℓ . The second difficulty is that although we want to classify the fibre into good and bad just like in Lemma 14, we have to face

the case that the fibres have height too small to be classified when ℓ and $t_2 \cdots t_s$ are large. Hence we need to split the sum into three pieces, and estimate the first two pieces where the height is small by the divisor bounds of Lemma 4. The third piece is where the height is large, and we need to reparameterise the base for this piece to reduce to Lemma 14.

Proof. **Step 1 (fibre sums).** We start from

$$I_1 = S_g(\mathcal{F}_I^\circ) = \sum_{(n_1, \dots, n_s, m_1, \dots, m_s) \in \mathcal{F}_I^\circ} \prod_{i=1}^s e(g(n_i) - g(m_i)).$$

We use Lemma 18 to parameterise the elements of $\mathcal{F}_I^\circ$. Let $\mathcal{P} = \pi_2(\psi(\mathcal{F}_I^\circ))$ be the finite set of outer parameters, as above. By Lemma 18, each $v \in \mathcal{P}$ has the form

$$v = (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s).$$

In Figure 4 this is the base after projection. For $v \in \mathcal{P}$ put

$$T(v) := t_2 \cdots t_s, \quad Q(v) := m'_2 \cdots m'_s, \quad M(v) := \max\{n'_1, m'_1\}.$$

Lemma 18 gives the fibre with $\gcd(b, m'_j) = 1$ for $2 \leq j \leq s$. Since $\gcd(b, Q(v)) = 1$ if and only if $\gcd(b, m'_j) = 1$ for $2 \leq j \leq s$, we may write

$$\mathcal{G}(v) = \left\{ (b, v) : \frac{A}{T(v)} \leq b \leq \frac{N}{T(v)M(v)}, \quad \gcd(b, Q(v)) = 1 \right\}.$$

The identities of Lemma 18 write the coordinates in terms of (b, v) , and the phase becomes

$$\prod_{i=1}^s e(g(n_i) - g(m_i)) = \Omega(v) e(g(bT(v)n'_1) - g(bT(v)m'_1)),$$

where $\Omega(v) := \prod_{i=2}^s e(g(n_i) - g(t_i m'_i))$ is unimodular and independent of b . Substituting into I_1 therefore gives

$$I_1 = \sum_{v \in \mathcal{P}} \Omega(v) S(v), \quad S(v) := \sum_{(b, v) \in \mathcal{G}(v)} e(g(bT(v)n'_1) - g(bT(v)m'_1)). \quad (6)$$

The inner sum $S(v)$ is the exponential sum over the fibre above the base point v . The fibre is exactly an interval with some additional coprime constraints, and the coprimality $\gcd(b, Q(v)) = 1$ is removed by Möbius inversion by adding a new parameter in Step 2.

Step 2 (Möbius inversion). In this step, our goal is to remove the coprime constraints by introducing a new parameter ℓ . Write

$$\mathcal{B}(v) := \left\{ b \in \mathbb{Z}_{>0} : \frac{A}{T(v)} \leq b \leq \frac{N}{T(v)M(v)} \right\}$$

for the interval without the coprimality constraint, so $\mathcal{G}(v) = \{(b, v) : b \in \mathcal{B}(v), \gcd(b, Q(v)) = 1\}$. The coprimality indicator is

$$\mathbf{1}_{\gcd(b, Q(v))=1} = \sum_{\ell | \gcd(b, Q(v))} \mu(\ell),$$

where $\ell \mid \gcd(b, Q(\nu))$ if and only if $\ell \mid Q(\nu)$ and $\ell \mid b$. Inserting the identity into the constrained sum over $\mathcal{B}(\nu)$ and changing the order of two finite sums therefore gives, for any function H on $\mathbb{Z}_{>0}$,

$$\sum_{\substack{b \in \mathcal{B}(\nu) \\ \gcd(b, Q(\nu))=1}} H(b) = \sum_{\ell \mid Q(\nu)} \mu(\ell) \sum_{\substack{k \geq 1 \\ \ell k \in \mathcal{B}(\nu)}} H(\ell k). \quad (7)$$

The fibre is rewritten as an interval in k of height $N/(\ell T(\nu)M(\nu))$. Apply (7) with $H(b) = e(g(bT(\nu)n'_1) - g(bT(\nu)m'_1))$. Then

$$S(\nu) = \sum_{\ell \mid Q(\nu)} \mu(\ell) S(\nu, \ell), \quad (8)$$

where

$$S(\nu, \ell) := \sum_{\substack{k \geq 1 \\ \frac{A}{\ell T(\nu)} \leq k \leq \frac{N}{\ell T(\nu)M(\nu)}}} e(g(\ell k T(\nu)n'_1) - g(\ell k T(\nu)m'_1))$$

Taking absolute values in (6)–(8), and using $|\mu(\ell)| \leq 1$ together with $|\Omega(\nu)| = 1$,

$$|I_1| \leq \sum_{\nu \in \mathcal{P}} \sum_{\ell \mid Q(\nu)} |S(\nu, \ell)|. \quad (9)$$

For the form of $S(\nu, \ell)$, we view it as the exponential sum over a fibre, as drawn in Figure 5. In the same spirit as in the proof of Lemma 14, we want to classify these fibres into good and bad. But when we look at the height of the fibre, that is, $N/\ell T(\nu)M(\nu)$, with ℓ and ν fixed. These additional parameters ℓ and $T(\nu)$ will cause trouble when we try to copy the same argument as in Lemma 14, since when ℓ and $T(\nu)$ are large, the height of the fibre is too small to be an input of Lemma 10. Our solution, as shown in the next step, is to isolate those large ℓ and ν with $T(\nu)$ large, and then apply the same argument as in Lemma 14 to bound the rest.

Step 3 (three-piece split). As noted above, we need to isolate those large ℓ and ν with $T(\nu)$ large, and then apply the same argument as in Lemma 14 to bound the rest. Partition the pairs (ν, ℓ) in (9) according to the size of $T(\nu)$ and ℓ , and write

$$\begin{aligned} \Xi_I &:= \sum_{\substack{\nu \in \mathcal{P} \\ T(\nu) \geq \delta^{-1}}} \sum_{\ell \mid Q(\nu)} |S(\nu, \ell)|, \\ \Xi_{II} &:= \sum_{\substack{\nu \in \mathcal{P} \\ \ell \mid Q(\nu) \\ \ell \geq \delta^{-1}}} |S(\nu, \ell)|, \\ \Xi_{III} &:= \sum_{\substack{\nu \in \mathcal{P} \\ T(\nu) < \delta^{-1}}} \sum_{\substack{\ell \mid Q(\nu) \\ \ell < \delta^{-1}}} |S(\nu, \ell)|. \end{aligned}$$

Piece I is the contribution of ν with large $T(\nu)$. Piece II is the contribution of large ℓ . Piece III is what remains when both $T(\nu)$ and ℓ are small. There is a harmless double counting when both $T(\nu)$ and ℓ are at least δ^{-1} . Hence

$$|I_1| \leq \Xi_I + \Xi_{II} + \Xi_{III}.$$

Pieces I and II are estimated by the divisor bounds of Lemma 4 in this step; The third piece contains the remaining base points with small $T(\nu)$ and ℓ , where those base points correspond to fibres of height large enough to be classified in the next step.

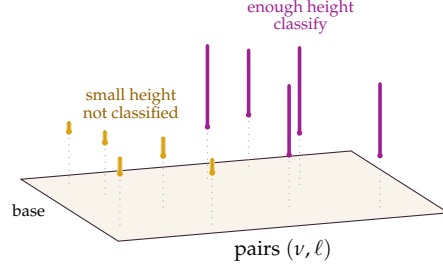


Figure 5: Each fibre is the floating interval. Gold: small height, excluded from classification (Pieces I and II). Purple: enough height to enter the good/bad classification of Lemma 14 (Piece III).

Figure 5 records this isolation.

Next, we deal with Piece I and II.

Piece I ($T(\nu) \geq \delta^{-1}$). Recall that each $\nu \in \mathcal{P}$ is

$$\nu = (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s), \quad (10)$$

and write $T(\nu) = t_2 \cdots t_s$, $Q(\nu) = m'_2 \cdots m'_s$.

Piece I is dealt with by the divisor bounds of Lemma 4. We do not use exponential sum bounds here. The sum $S(\nu, \ell)$ runs over those k with $\ell k \in \mathcal{B}(\nu)$, and each term has absolute value 1. We use the trivial bound. Hence

$$\Xi_I \leq \sum_{\substack{\nu \in \mathcal{P} \\ T(\nu) \geq \delta^{-1}}} \sum_{\ell | Q(\nu)} \sum_{\substack{k \geq 1 \\ \ell k \in \mathcal{B}(\nu)}} 1.$$

We need to bound the right hand side of the above, while the only budget we have is that the sum is over large $T(\nu)$. Our strategy is to change the order of summation and count the number of ν with fixed T, ℓ, k . Changing the order, the right hand side of the above equals

$$\sum_{T \geq \delta^{-1}} \sum_{\ell \geq 1} \sum_{k \geq 1} \#\{\nu \in \mathcal{P} : T(\nu) = T, \ell \mid Q(\nu), \ell k \in \mathcal{B}(\nu)\}.$$

The above is saying that we first fix T, ℓ, k and count the number of base points ν that satisfy the conditions. The observation is that the number of ν with large $T(\nu)$ is well bounded.

Claim 2. *Let $T, \ell, k \geq 1$. Then*

$$\#\{\nu \in \mathcal{P} : T(\nu) = T, \ell \mid Q(\nu), \ell k \in \mathcal{B}(\nu)\} \leq \frac{N^s}{k \ell^2 T^2} \tau_{s-1}(T) \tau_{s-1}(\ell) \tau_s(\ell T) (2s \log N)^{s^2-1}. \quad (11)$$

Proof of the claim. Recall that we write ν as in (10). For fixed T, ℓ, k , we first observe the conditions that ν in the above set must satisfy, and then use the conditions to bound the number of ν with fixed T, ℓ, k . The conditions $T(\nu) = T$ and $\ell \mid Q(\nu)$ imply

$$n'_1 n_2 \cdots n_s = T m'_1 m'_2 \cdots m'_s, \quad \ell \mid m'_2 \cdots m'_s, \quad (12)$$

while $\ell k \in \mathcal{B}(\nu)$ implies $\ell k T n'_1 \leq N$ and $\ell k T m'_1 \leq N$. Also note that $n_2, \dots, n_s \leq N$. Gathering conditions we need, it therefore suffices to bound

$$\#\{(t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, \dots, m'_s) : t_2 \cdots t_s = T, (12) \text{ holds}, n_2, \dots, n_s \leq N, \ell k T n'_1 \leq N\}.$$

Our strategy is to first fix the following product

$$n := n'_1 n_2 \cdots n_s = T m'_1 m'_2 \cdots m'_s,$$

and bound the number of tuples for this n by the divisor function, and then do the summation over all possible values of n .

We first observe the possible range of n . Since $n_2, \dots, n_s \leq N$ and $n'_1 \leq N/(\ell k T)$, the possible n satisfy $n \leq N^s/(\ell k T)$. From $n = T m'_1 m'_2 \cdots m'_s$ and $\ell \mid m'_2 \cdots m'_s$, one has $\ell T \mid n$.

Now fix such an n . The tuple $(t_2, \dots, t_s)$ is an ordered $(s-1)$ -factorisation of the already fixed product T , hence at most $\tau_{s-1}(T)$ choices. The condition $\ell \mid Q(v)$ does not specify how ℓ sits inside $m'_2, \dots, m'_s$. Write $\ell = \ell_2 \cdots \ell_s$ with $\ell_i \mid m'_i$ and $m'_i = \ell_i r_i$. There are at most $\tau_{s-1}(\ell)$ such allocations. After this substitution the identity $n = T m'_1 m'_2 \cdots m'_s$ becomes $n = \ell T m'_1 r_2 \cdots r_s$. Thus $(m'_1, r_2, \dots, r_s)$ is an ordered s -factorisation of $n/(\ell T)$, hence at most $\tau_s(n/(\ell T))$ choices. Independently, $(n'_1, n_2, \dots, n_s)$ is an ordered s -factorisation of n , hence at most $\tau_s(n)$ choices. For this n , the number of tuples is therefore at most $\tau_{s-1}(T) \tau_{s-1}(\ell) \tau_s(n) \tau_s(n/(\ell T))$.

Summing over all possible values of n gives

$$\tau_{s-1}(T) \tau_{s-1}(\ell) \sum_{\substack{n \leq N^s/(\ell k T) \\ \ell T \mid n}} \tau_s(n) \tau_s\left(\frac{n}{\ell T}\right).$$

Write $n = \ell T m$ and use $\tau_s(\ell T m) \leq \tau_s(\ell T) \tau_s(m)$. Then

$$\sum_{\substack{n \leq N^s/(\ell k T) \\ \ell T \mid n}} \tau_s(n) \tau_s\left(\frac{n}{\ell T}\right) \leq \tau_s(\ell T) \sum_{m \leq N^s/(k \ell^2 T^2)} \tau_s(m)^2,$$

and Lemma 4 gives the asserted bound. This completes the proof of the claim. $\square$

Now sum (11) over T , ℓ and k . Here $T \leq N^{s-1}$ and, since $\ell \mid Q$, also $\ell \leq N^{s-1}$. Using $\sum_{k \leq N} k^{-1} \ll \log N$ and $\tau_s(\ell T) \leq \tau_s(\ell) \tau_s(T)$, Ξ_I is at most

$$N^s (2s \log N)^{s^2} \left(\sum_{\ell \leq N^{s-1}} \frac{\tau_s(\ell) \tau_{s-1}(\ell)}{\ell^2} \right) \left(\sum_{\delta^{-1} \leq T \leq N^{s-1}} \frac{\tau_s(T) \tau_{s-1}(T)}{T^2} \right).$$

Since $\tau_{s-1} \leq \tau_s$, Lemma 4 applies to both sums. The ℓ -sum is a full sum and the T -sum is a tail from δ^{-1} , so this is

$$\ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

Piece II ($\ell \geq \delta^{-1}$). The bound (11) is symmetric in the final T - and ℓ -sums. This time the T -sum is unrestricted and Lemma 4 is applied to the tail $\ell \geq \delta^{-1}$. Hence

$$\Xi_{II} \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

Step 4 (reparameterisation of the base; $T(v), \ell < \delta^{-1}$). It remains to bound Ξ_{III} . Both $T(v)$ and ℓ are now small. Hence our strategy is the same as in Lemma 14, that is, to conduct exponential sums over the fibres, group those base points as good or bad, and then apply the same argument as in Lemma 14 to bound the bad base points. In this step, a reparameterisation of the base is needed. Recall that Piece III is

$$\Xi_{III} = \sum_{\substack{v \in \mathcal{P} \\ T(v) < \delta^{-1}}} \sum_{\substack{\ell \mid Q(v) \\ \ell < \delta^{-1}}} |S(v, \ell)|.$$

Recall that each $v \in \mathcal{P}$ is a $(3s - 1)$ -dimensional integral vector

$$v = (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s).$$

For each v , we write $T(v) = t_2 \cdots t_s$, $Q(v) = m'_2 \cdots m'_s$, and $M(v) = \max\{n'_1, m'_1\}$. Note that for each v fixed, and ℓ fixed, considering the inner sum $S(v, \ell)$, we have

$$S(v, \ell) = \sum_{\frac{A}{\ell T(v)} \leq k \leq \frac{N}{\ell T(v) M(v)}} e(g(\ell k T(v) n'_1) - g(\ell k T(v) m'_1)).$$

This is the desired exponential sum over an arithmetic progression. Write $h = \ell T(v)$. Our plan is to apply Lemma 10 to the polynomial $g_h(\cdot)$ defined by $g_h(\cdot) = g(\cdot h)$. This requires the outer parameters $(t_2, \dots, t_s)$ and ℓ to be fixed first.

Let $\pi_T(v) = (t_2, \dots, t_s)$ be the projection of v onto the first $s - 1$ coordinates, and let $\mathcal{P}_T := \pi_T(\mathcal{P})$ be the set of images of $\mathcal{P}$ under this projection. As mentioned above, we may first fix the $(t_2, \dots, t_s)$ coordinates and ℓ and sum over the remaining coordinates to use the same argument as in Lemma 14. So this requires us to change the order of summation.

Changing the order of summation, and writing $\mathbf{t} = (t_2, \dots, t_s)$, we obtain

$$\Xi_{\text{III}} = \sum_{\substack{\mathbf{t} \in \mathcal{P}_T \\ t_2 \cdots t_s < \delta^{-1}}} \sum_{\ell < \delta^{-1}} \sum_{\substack{v \in \mathcal{P} \\ \pi_T(v) = \mathbf{t} \\ \ell | Q(v)}} |S(v, \ell)|.$$

For $(\mathbf{t}, \ell)$ fixed, and hence $T = t_2 \cdots t_s$ and $h = \ell T$ fixed, we consider the sum over the fibres and the base points v .

$$\sum_{\substack{v \in \mathcal{P} \\ \pi_T(v) = \mathbf{t}}} |S(v, \ell)|. \quad (13)$$

Our plan is to repeat the same argument as in Lemma 14 to bound the bad base points. To get the same form as in Lemma 14, we need to reparameterise the base. For $\mathbf{t} = (t_2, \dots, t_s)$ fixed, consider all $v = (t_2, \dots, t_s, n'_1, n_2, \dots, n_s, m'_1, m'_2, \dots, m'_s) \in \mathcal{P}$. We record the conditions that v must satisfy in order to put the base in the same form as in Lemma 12:

- (i) $n'_1 n_2 \cdots n_s = m'_1 (t_2 m'_2) \cdots (t_s m'_s)$
- (ii) $\gcd(n'_1, m'_1) = 1$ and $n'_1 \neq m'_1$
- (iii) $n'_1, \dots, n_s, m'_1 \leq N$
- (iv) $t_j m'_j \leq N$ for $2 \leq j \leq s$.

Conditions (i) and (ii) come from Lemma 18, and (i) is the same product equation as in Remark 13. Conditions (iii) and (iv) are the bounds coming from V . Every v in (13) satisfies (i)–(iv), so summing instead over all positive integers $n'_1, \dots, n_s, m'_1, \dots, m'_s$ obeying these conditions only enlarges the range:

$$(13) \leq \sum_{(i)-(iv)} \left| \sum_{A/h \leq k \leq N/(h \max\{n'_1, m'_1\})} e(g(hkn'_1) - g(hkm'_1)) \right|.$$

The inner sum does not depend on $m'_2, \dots, m'_s$. For $\mathbf{t}$ fixed, set $m_j = t_j m'_j$ when $2 \leq j \leq s$, and rename n'_1, m'_1 to n_1, m_1 . The product equation (i) becomes $n_1 \cdots n_s = m_1 \cdots m_s$, with conditions (iii) and (iv) turned into $m_j, n_j \leq N$ when $1 \leq j \leq s$ and $t_j \mid m_j$ when $2 \leq j \leq s$. Dropping the divisibility $t_j \mid m_j$ only enlarges the range. Applying (ii), the previous display is at most the same sum over

$$\mathcal{V} := \{(\vec{n}_s, \vec{m}_s) \in V : \gcd(n_1, m_1) = 1, n_1 \neq m_1\}.$$

Thus the previous bound is at most

$$\sum_{(\vec{n}_s, \vec{m}_s) \in \mathcal{V}} \left| \sum_{\frac{A}{h} \leq k \leq \frac{N}{h \max\{n_1, m_1\}}} e(g(hkn_1) - g(hkm_1)) \right|.$$

By Remark 13 we may apply the map ϕ from Lemma 12 to $\mathcal{V}$ and get the parameterisation $(a_{ij})_{1 \leq i, j \leq s}$ for each $(\vec{n}_s, \vec{m}_s) \in \mathcal{V}$. Since $\gcd(n_1, m_1) = 1$ for every point of $\mathcal{V}$, the construction gives $a_{11} = 1$. As in Lemma 12, write $A_1 = \prod_{j=2}^s a_{1j}$, $B_1 = \prod_{i=2}^s a_{i1}$, and $M_1 = \max\{A_1, B_1\}$. Since $a_{11} = 1$, one has $n_1 = A_1$, $m_1 = B_1$, and $\max\{n_1, m_1\} = M_1$, so the summation above becomes

$$\sum_{\phi(\mathcal{V})} \left| \sum_{A/h \leq k \leq N/(hM_1)} e(g(hkA_1) - g(hkB_1)) \right|.$$

Moreover, the inner sum is constant along the remaining diagonal $(a_{22}, \dots, a_{ss})$, and that fibre has at most $\prod_{i=2}^s N/M_i$ points, the same trivial bound as in (3). Summing over all off-diagonal matrices therefore yields

$$\sum_{\pi_1(\phi(\mathcal{V}))} \left| \sum_{A/h \leq k \leq N/(hM_1)} e(g(hkA_1) - g(hkB_1)) \right| \prod_{i=2}^s \frac{N}{M_i}. \quad (14)$$

This is the scaled form of (3), where the main differences from (3) are that the polynomial $g(\cdot)$ is replaced by $g(h \cdot)$ and the fibre range is divided by a factor h .

Apply the same arguments as in Lemma 14 to $g_h(n) := g(hn)$. The summation range is $A/h \leq k \leq N/(hM_1)$. The conditions required for the argument are guaranteed by $h = \ell T < \delta^{-2}$ and the restricted range of A . Hence, unless alternative (2) occurs for g_h , the argument of Claim 1 applies with N and A replaced by N/h and A/h , respectively. The same estimates for Σ_I and Σ_{II} as in Lemma 14 apply, with the factor N/M_1 replaced by $N/(hM_1)$. Both sums therefore pick up an extra $1/h$. Thus, unless the Diophantine alternative for g_h occurs,

$$(14) \ll_{d,s} \delta^{1/2} \frac{N^s}{h} (\log N)^{O(s^2)}. \quad (15)$$

Suppose alternative (2) occurs for g_h . Then there is a nonzero integer K with $|K| \ll_d \delta^{-O_d(1)}$ and $\|Kg_h\|_{C^\infty[N/h]} \ll_d \delta^{-O_d(1)}$. Write $g(n) = \sum_{j=0}^d \beta_j n^j$, so $g_h(n) = \sum_{j=0}^d (\beta_j h^j) n^j$. Set $K' = Kh^d$. Since $h = \ell T < \delta^{-2}$, one has $|K'| \ll_d \delta^{-O_d(1)}$. By a simple calculation,

$$\|K'g\|_{C^\infty[N]} \leq h^d \|Kg_h\|_{C^\infty[N/h]} \ll_d \delta^{-O_d(1)}.$$

Thus K' is the integer in alternative (2) for g .

So far we have bounded the sum over the fibres and the base points ν with $\mathbf{t}$ and ℓ fixed, that is, the quantity (13). Finally, we need to bound the sum over all $\mathbf{t}$ and ℓ . The number of vectors $\mathbf{t}$ with product T is at most $\tau_{s-1}(T)$, while ℓ is already a scalar. Recall that $h = \ell T$. Summing (15) gives the additional factor

$$\left(\sum_{T < \delta^{-1}} \frac{\tau_{s-1}(T)}{T} \right) \left(\sum_{\ell < \delta^{-1}} \frac{1}{\ell} \right) \ll (2 \log N)^s$$

by Lemma 4. Multiplying the right hand side of (15) by $(2 \log N)^s$ remains of size $\delta^{1/2} N^s (\log N)^{O(s^2)}$. Hence

$$\Xi_{\text{III}} \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}.$$

Combining the three estimates, we obtain

$$|I_1| \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)},$$

unless Lemma 10 produces alternative (2). □

6.2 Diagonal part I_2

We now deal with I_2 . The following lemma is the goal of this section.

Lemma 20 (Diagonal I_2). *Let the parameters be as in Proposition 17. Recall that, with $i_1 = 1$,*

$$I_2 = S_g(\mathcal{F}_{i_1} \cap \cdots \cap \mathcal{F}_{i_k} \cap \{n_1 = m_1\}).$$

Then

$$|I_2| \ll A^{-1} N^s (2 \log N)^{2s^2-2}.$$

In particular, $|I_2| \ll \delta N^s (2 \log N)^{2s^2-2} \ll_{d,s} \delta^{1/2} N^s (\log N)^{O(s^2)}$ once $A > \delta^{-1}$.

Proof. Write $r = i_2 \geq 2$. Membership in $\mathcal{F}_1$ on the diagonal $n_1 = m_1$ gives $n_1 = m_1 \geq A$. Membership in $\mathcal{F}_r$ gives $\gcd(n_1, m_r) \geq A$. Since $n_1 = m_1$, this is $\gcd(m_1, m_r) \geq A$. Thus the support of I_2 sits inside

$$\{n_1 = m_1, \gcd(m_1, m_r) \geq A\} \cap V.$$

Since $|e| = 1$, it is enough to count the cardinality of the above set.

Write $n = n_1 \cdots n_s = m_1 \cdots m_s \leq N^s$. We first fix this n and count the number of points satisfying the above conditions. Fix $h = \gcd(m_1, m_r)$ with $h \geq A$. Set $m_1 = ha$ and $m_r = hb$ with $\gcd(a, b) = 1$. Then $h \mid m_1$ and $h \mid m_r$ contribute two copies of h to the same factorisation of n , hence $h^2 \mid n$. The tuple $(a, b, (m_i)_{i \neq 1, r})$ is an ordered s -factorisation of n/h^2 , so the number of admissible $(m_1, \dots, m_s)$ is at most $\tau_s(n/h^2)$. The number of admissible $(n_1, \dots, n_s)$ is at most $\tau_s(n)$. Hence the contribution of this h is $\sum_{n \leq N^s} \tau_s(n/h^2) \tau_s(n)$. By Lemma 4 and multiplicativity,

$$\sum_{n \leq N^s} \tau_s(n/h^2) \tau_s(n) \ll \frac{\tau_s(h)^2}{h^2} N^s (2 \log N)^{s^2-1}.$$

Summing over $h \geq A$ via $\sum_{h \geq A} \tau_s(h)^2 / h^2 \ll A^{-1} (2 \log N)^{s^2-1}$ yields the claim. □

Combining Lemmas 19 and 20 proves Proposition 17.

7 Conclusion of the proof

Proof of Theorem 2. We induct on $s \geq 2$. The case $s = 2$ uses $\mathcal{H}(1)$. For $s \geq 3$, assume $\mathcal{H}(s-1)$.

Let $C_d > 5$ be as in Proposition 11, and choose $c_0 = c_0(d, s) > 0$ sufficiently small. Take N large enough depending on d and s that every estimate below applies. Fix $N^{-c_0} < \delta < 1/8$ and an integer A with

$$\delta^{-(C_d+2)} < A < \delta^{-(C_d+3)}.$$

Let g be a polynomial of degree d .

If alternative (2) occurs in Proposition 11 or Proposition 17, the same q gives alternative (2) of the theorem. Otherwise alternative (1) holds in both, and the rest of the proof combines those bounds.

By Lemma 5, $U_s(N) = S_g(V)$. Lemma 6 covers V by $\bigcup_i \mathcal{F}_i$ up to a complement of size $O(N^{s-1}A^{2s}(\log N)^{O(s^2)})$. The upper bound $A < \delta^{-(C_d+3)}$ together with $\delta > N^{-c_0}$ and c_0 small puts this complement inside $\delta^{1/2}N^s(\log N)^{O(s^2)}$.

Expand the union by (1). Proposition 11 and Remark 16 give

$$S_g(\mathcal{F}_i) = (s-1)!N^s + O_{d,s}(\delta^{1/2}N^s(\log N)^{O(s^2)})$$

for each i . For $\kappa \geq 2$, Proposition 17 bounds each κ -fold intersection by $O_{d,s}(\delta^{1/2}N^s(\log N)^{O(s^2)})$.

There are s single sets and at most 2^s intersection terms. Absorbing these factors into the implied constant,

$$U_s(N) = s!N^s + O_{d,s}(\delta^{1/2}N^s(\log N)^{O(s^2)}).$$

Dividing by N^s gives

$$|\mathcal{M}_s(N) - s!| \ll_{d,s} \delta^{1/2}(\log N)^{O(s^2)}.$$

Replacing the free parameter δ by δ^2 (and shrinking c_0) yields $\mathcal{H}(s)$. The Diophantine alternative becomes $|q| \ll_d (\delta^2)^{-O_d(1)} = \delta^{-O_d(1)}$, as stated. $\square$

Proof of Corollary 3. Fix an integer $s \geq 2$ and set

$$\delta = \delta_s(N) = (\log N)^{-K}, \tag{16}$$

where $K = K(s)$ is large enough depending only on s that the error $\delta(\log N)^{O(s^2)}$ in Theorem 2 tends to 0. For N large enough (depending on d and s), one has $\delta > N^{-c_0}$ with $c_0 = c_0(d, s)$ as in Theorem 2, so the theorem applies. Suppose towards a contradiction that alternative (2) holds for this δ : there exists $q \in \mathbb{Z} \setminus \{0\}$ with

$$|q| \ll_d \delta^{-O_d(1)} \quad \text{and} \quad \|qg\|_{C^\infty[N]} \ll_d \delta^{-O_d(1)}.$$

By Definition 1, the second bound means that for every $1 \leq i \leq d$,

$$\|q\beta_i\|_{\mathbb{R}/\mathbb{Z}} \ll_d \delta^{-O_d(1)} N^{-i}.$$

Since $|q| \ll_d \delta^{-O_d(1)}$ and $\delta = (\log N)^{-K}$ with $K = K(s)$, one has $|q| \ll_{d,s} (\log N)^{O_{d,s}(1)}$. Choose $\varepsilon = \varepsilon(d, s) > 0$ small enough that $|q|^\varepsilon \leq (\log N)/2$. The hypothesis then yields $C = C(g, \varepsilon) > 0$ such that some i has $\|q\beta_i\| \geq C \exp(-|q|^\varepsilon)$. Combining with the display above,

$$\exp(-|q|^\varepsilon) \ll_d \delta^{-O_d(1)} N^{-i}.$$

The left side is at least $N^{-1/2}$, while the right side is $\ll_{d,s} (\log N)^{O_{d,s}(1)} N^{-1}$. This is impossible for large N . Thus alternative (2) cannot occur, and Theorem 2 gives

$$|\mathbb{E}|S_N|^{2s} - s!| \ll_{d,s} \delta(\log N)^{O(s^2)} \rightarrow 0$$

as $N \rightarrow \infty$. Thus $\mathcal{M}_s(N) = \mathbb{E}|S_N|^{2s} \rightarrow s!$ for every fixed integer $s \geq 2$. (The case $s = 1$ is immediate: $\mathbb{E}|S_N|^2 = 1$ by Steinhaus orthogonality and $|e(g(n))| = 1$.)

For odd moments, fix integers $s_1, s_2 \geq 0$ with $s_1 \neq s_2$; by conjugation, we may assume $s_2 < s_1$. If $s_2 = 0$, then the Steinhaus orthogonality gives

$$\mathbb{E}[S_N^{s_1}] = N^{-s_1/2} e(s_1 g(1)) \rightarrow 0.$$

We may therefore assume that $1 \leq s_2 < s_1$. Expanding and using complete multiplicativity together with the Steinhaus orthogonality relation, we obtain

$$\mathbb{E}[S_N^{s_1} \overline{S_N^{s_2}}] = N^{-(s_1+s_2)/2} \sum_{\substack{n_1, \dots, n_{s_1} \leq N \\ m_1, \dots, m_{s_2} \leq N \\ n_1 \cdots n_{s_1} = m_1 \cdots m_{s_2}}} \prod_{i=1}^{s_1} e(g(n_i)) \prod_{j=1}^{s_2} \overline{e(g(m_j))}.$$

Since $|e(g(\cdot))| = 1$, the absolute value of the right-hand side is at most

$$N^{-(s_1+s_2)/2} \cdot \#\{(n_1, \dots, n_{s_1}, m_1, \dots, m_{s_2}) \in [1, N]^{s_1+s_2} : n_1 \cdots n_{s_1} = m_1 \cdots m_{s_2}\}.$$

Write $P = n_1 \cdots n_{s_1} = m_1 \cdots m_{s_2}$. Then $P \leq N^{s_2}$, and for each such P the number of ordered factorisations is at most $\tau_{s_1}(P) \tau_{s_2}(P)$. Hence the cardinality is

$$\leq \sum_{P \leq N^{s_2}} \tau_{s_1}(P) \tau_{s_2}(P).$$

Cauchy–Schwarz and Lemma 4 give

$$\begin{aligned} \sum_{P \leq N^{s_2}} \tau_{s_1}(P) \tau_{s_2}(P) &\leq \left(\sum_{P \leq N^{s_2}} \tau_{s_1}(P)^2 \right)^{1/2} \left(\sum_{P \leq N^{s_2}} \tau_{s_2}(P)^2 \right)^{1/2} \\ &\leq N^{s_2} (2 \log N)^{O((s_1+s_2)^2)}. \end{aligned}$$

Therefore

$$|\mathbb{E}[S_N^{s_1} \overline{S_N^{s_2}}]| \ll_{s_1, s_2} N^{(s_2-s_1)/2} (\log N)^{O((s_1+s_2)^2)} \rightarrow 0 \quad (N \rightarrow \infty).$$

Thus $\mathbb{E}[S_N^{s_1} \overline{S_N^{s_2}}]$ tends to 0 whenever $s_1 \neq s_2$. Together with $\mathbb{E}|S_N|^{2s}$ tending to $s!$ for every $s \geq 1$, the moments of S_N match those of a standard complex normal with mean 0 and variance 1. By the moment method [4, Chapter 5, Theorem 8.6], S_N converges in law to this distribution. $\square$

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